

Project Data Page:





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DESIGN CALCULATION

In Accordance with ASME Section VIII Division 1

ASME Code Version : 2025

Analysis Performed by : SL User (Modify in Config->Save Options)

Job File : ...\\406-V-0067%.pvdb

Date of Analysis : Jul 22,2026 9:56am

PV Elite 28, January 2026

Vessel Design Summary:

ASME Code, Section VIII Division 1, 2025

Diameter Spec : 600.000 mm ID
 Vessel Design Length, Tangent to Tangent 806.18 mm
 Distance of Bottom Tangent above Grade 300.00 mm
 Specified Datum Line Distance 0.00 mm
 Internal Design Temperature 65 °C
 Internal Design Pressure 0.100 MPa
 External Design Temperature 65 °C
 External Design Pressure 0.001 MPa
 Maximum Allowable Working Pressure 0 MPa
 External Max. Allowable Working Pressure 1 MPa
 Shop Test Pressure 0.140 MPa
 Field Test Pressure 0.184 MPa
 Required Minimum Design Metal Temperature 11.4 °C
 Warmest Computed Minimum Design Metal Temperature -37.0 °C
 Wind Design Code IS 875 (PART 3) Amd. 1&2 (2020)
 Earthquake Design Code IS 1893 RSM Amd. 1&2 (2018)

Materials of Construction:

Component Type	Material	Class	Thickness	UNS #	Material Treatment Options
Shell	SA-516 70	...	...	K02700	
Cover	SA-516 70	...	...	K02700	
Flange	SA-516 70	...	...	K02700	
Nozzle	SA-106 B	...	...	K03006	
Nozzle	SA-312 TP304L	...	...	S30403	
Re-Pad	SA-516 70	...	...	K02700	
Nozzle Flg	SA-105	...	...	K03504	
Nozzle Flg	SA-182 F304L	...	> 130	S30403	
Flg Bolting	SA-193 B7	...	<= 64	G41400	
Leg Baseplate		...	...		
Leg Bolting	SA-193 B7	...	<= 64	G41400	

+FG - Fine Grained, +R - rolled, +T - Tempered, +N - Normalized
 +Q - quenched, +CBV - Cb & V added, +IT - Impact tested

Element Pressures and MAWP (MPa & mm):

Element Description or Type	Design Pressure + Stat. head	Ext. Press.	Element M.A.W.P	Total Corrosion Allowance	Str. Flg. Gov.	In Creep Range
Bottom Head	0.109	0.00	0.000	6.0000	N/A	No
Shell-1	0.109	0.00	2.000	6.0000	N/A	No
Body Flange-1	0.100	0.00	2.000	6.0000	N/A	No
Body Flange-2	0.100	0.00	1.000	6.0000	N/A	No

Liquid Level: 500.00 mm Dens.: 0.002 kg/cm³ Sp. Gr.: 1.864

Element Types and Properties:

Element Type	"To" Elev mm	Element Length mm	Nominal Thickness mm	Finished Thickness mm	Reqd Thk Internal mm	Reqd Thk External mm	Long Eff	Circ Eff
Wld Flat	18.0	18.0	18.0	18.0	16.2	...	0.85	0.85



Cylinder	675.0	657.0	10.0	10.0	7.5	6.2	0.85	0.85
Body Flg	768.2	90.0	75.0	75.0	72.4	72.4	1.00	1.00
Body Flg	806.2	38.0	38.0	38.0	35.2	35.3	1.00	1.00

Local Stress Analysis Results:

Description	Analysis Type	Max Stress Ratio	High Stress Location	Pass Fail
N1_4"	WRC-297	0.756	Shell	Passed
N2_6"	WRC-297	0.964	Nozzle	Passed
N3_2"	WRC-297	0.558	Shell	Passed

Loads for Foundation/Support Design:

Total Wind Shear on top of all Legs	920.	N
Total Earthquake Shear on top of all Legs	738.	N
Total Wind Moment at top of all Legs	293721.	N-mm
Total Earthquake Moment at top of all Legs	288962.	N-mm
Wind Load Case Test Moment on Support	116646.	N-mm
Earthquake Load Case Test Moment on Support	29430.	N-mm
Test Case Wind Shear on Support	303.	N
Max. Wind Shear on one Leg (top & bottom)	607.	N
Max. Earthq. Shear on one Leg (top & bottom)	488.	N
Max. Wind Moment at base of one Leg	443430.	N-mm
Max. Earthquake Moment at base of one Leg	356020.	N-mm
Max. Vertical Load (Wt. + Wind) on one Leg	3492.	N
Max. Vertical Load (Wt. + Eq.) on one Leg	3478.	N

Wind and Earthquake moments include the effects of user defined forces and moments if any exist in the job and were specified to act (compute loads and stresses) during these cases. Also included are moment effects due to eccentric weights if any are present in the input.

Weights:

Fabricated - Bare w/o removable internals	577.7	kgm
Shop Test - Fabricated + water (full)	789.7	kgm
Shipping - Fab. + rem. intls.+ shipping app.	577.7	kgm
Erected - Fab. + rem. intls.+ insul. (etc)	577.7	kgm
Empty - Fab. + intls. + details + wghts.	577.7	kgm
Operating - Empty + operating liquid (no ca)	841.1	kgm
Field Test - Empty weight + test medium (full)	789.7	kgm

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Nozzle Calculation Summary (MPa & mm):

Description	MAWP	Ext	MAPNC	UG-45	[tr]	Weld Path	Areas or Stresses
N1_4"	2	OK	...	OK	7.50	OK	Passed
N2_6"	2	OK	...	OK	7.50	OK	Passed
N3_2"	2	...	...	OK	7.50	OK	No Calc[*]
N4_2"	1	OK	...	OK	10.80	OK	Passed

Nozzle MAWP Summary:

Minimum MAWP Nozzles : 1 Nozzle : N42"
 Minimum MAWP Shells/Flanges : 0 Element : Bottom Head
 Minimum MAPnc Shells/Flanges : 0 Element : Bottom Head

Computed Vessel M.A.W.P. : 0 MPa

[*] - This was a small opening and the areas were not computed or the MAWP of this connection could not be computed because the longitudinal bending stress was greater than the hoop stress.

Note: MAWPs (Internal Case) shown above are at the High Point.

Multiple output lines for the same nozzle indicates required Code calculations in both the longitudinal and circumferential planes of reinforcement where applicable.

Check the spatial relationship between the nozzles:

From Node	Nozzle Description	Y Coordinate mm	Layout Angle deg	Dia. Limit mm
20	N1_4"	468.000	270.000	198.650
20	N2_6"	468.000	90.000	303.451
20	N3_2"	218.000	90.000	125.600
40	N4_2"	0.000	90.000	141.724

The nozzle spacing is computed by the following:

$$= \sqrt{ll^2 + lc^2} \text{ where}$$

ll - Arc length along the inside vessel surface in the long. direction.
 lc - Arc length along the inside vessel surface in the circ. direction

If any interferences/violations are found, they will be noted below.

No interference violations have been detected!

Checking Multiple Nozzles on Flat Head per ASME Sec. VIII Div. 1 UG-39

Comparing Nozzles on Element: Body Flange-2

Note: No Nozzle pairs found on this element.

UG-39 Nozzle Diameter and Distance to Edge Checks:

Nozzle Description	Nozzle dia. mm	Head Dia. /2 mm	Distance from Edge mm	Nozzle dia./4 mm
N4_2"	62.8	319.58	346.64	15.7

No Multiple Nozzle spacing violations detected!

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Nozzle Schedule:

Description	Nominal or Actual Size	Schd or FVC Type	Flg Type	Nozzle O/Dia in	Wall Thk mm	Reinforcing Pad Diameter	Pad Thk mm	Cut Length mm	Flg Class
N3_2"	2.000 in	Actual	LWN	3.060	13.462	178.00	10.00	241.95	300
N4_2"	2.000 in	Actual	LWN	3.060	13.462	100.00	38.00	267.48	300
N1_4"	4.000 in	160	WNF	4.500	13.487	...	...	135.17	300
N2_6"	6.000 in	120	WNF	6.625	14.275	...	...	140.66	300

General Notes for the above table:

The cut length is the outside projection + inside projection + drop + in-plane shell thickness. This value does not include weld gaps, nor does it account for shrinkage.

In the case of oblique nozzles, the outside diameter must be increased. The re-pad width around the nozzle is calculated as follows:

Width of Pad = (Pad Outside Dia. (per above) - Nozzle Outside Dia.)/2

For hub nozzles, the thickness and diameter shown are those of the smaller and thinner section.

Nozzle Material and Weld Fillet Leg Size Details (mm):

Description	Material	Shl Grve Weld	Noz Shl/Pad Weld	Pad OD Weld	Pad Grve Weld	Inside Weld
N3_2"	SA-106 B	10.000	10.000	8.000	10.000	...
N4_2"	SA-106 B	38.000	10.000	30.000	38.000	...
N1_4"	SA-106 B	10.000	10.000	...	...	...
N2_6"	SA-312 TP304L	10.000	10.000	...	...	...

Note: The Outside projections below do not include the flange thickness.

Nozzle Miscellaneous Data:

Description	Elev/Distance From Datum mm	Layout Angle deg	Proj Outside mm	Proj Inside mm	Installed in Component
N3_2"	218.000	90.0	229.43	0.00	Shell-1
N4_2"	...	90.0	229.43	0.00	Body Flange-2
N1_4"	468.000	270.0	115.67	0.00	Shell-1
N2_6"	468.000	90.0	112.62	0.00	Shell-1

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Bill of Materials:

QTY	DESCRIPTION	MATERIAL
1	FLAT HEAD: 18.0mm THK X 620.0mm OD	SA-516 70
1	WELD NECK FLANGE: 75.0mm THK X 771.0mm OD	SA-516 70
1	BLIND FLANGE: 38.0mm THK X 771.0mm OD	SA-516 70
3	LEGS, SHAPE: ISA75X75X8 730.0mm	...
3	BASEPLATES: 200.0mm X 200.0mm X 15.0mm	SA-516 70
3	BASEPLATE BOLTS: 20.0mm	SA-193 B7
3	LEG RE-PADS: 120.000mm X 120.000mm X 10.000mm	SA-516 70
1	GASKET: 652.0mm OD X 626.0mm ID	...
24	BODY FLANGE BOLTS: 24.0mm DIA	SA-193 B7
48	NUTS FOR BODY FLANGE BOLTS: 24.0mm DIA	...
2	CYLINDRICAL SEGMENT 10.0mm THK X 750.0mm X 657.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 10.0mm THK X 384.9mm X 657.0mm	SA-516 70
1	NAMEPLATE	...



Minimum Design Metal Temperature Results Summary (°C):

Description	Curve	Basic MDMT	Reduced MDMT	UG-20(f) MDMT	Thickness ratio	Gov Thk mm	E*	PWHT reqd
Notes								
Body Flange-1[11]	B	-29	-104	-29	0.044	10.000	1.00	No
Body Flange-2[11]	B	-29	-104	-29	0.117	9.500	1.00	No
Bottom Head [12]	B	-29	-37	-29	0.850	10.000	0.85	No
Shell-1 [8]	B	-29	-104	-29	0.319	10.000	0.85	No
N14" [1]	B	-3	-104	-29	0.007	11.801	1.00	No
Nozzle Flg [5]	A	-18	-104					
N32" [1]	B	-29	-104	-29	0.059	10.000	1.00	No
N42" [1]	B	-19	-104	-29	0.004	13.462	1.00	Yes
Nozzle Flg [5]	A	-18	-104					
Bolting [21]		-48						

Warmest MDMT: -3 -37

Required Minimum Design Metal Temperature 11.4 °C

Warmest Computed Minimum Design Metal Temperature -37.0 °C

Notes:

- [!] - This was an impact tested material.
- [1] - Governing Nozzle Weld.
- [4] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-c).
- [5] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-b).
- [6] - MDMT Calculations at the Shell/Head Joint.
- [7] - MDMT Calculations for the Straight Flange.
- [8] - Cylinder/Cone/Flange Junction MDMT.
- [9] - Calculations in the Spherical Portion of the Head.
- [10] - Calculations in the Knuckle Portion of the Head.
- [11] - Calculated (Body Flange) Flange MDMT.
- [12] - Calculated Flat Head MDMT per UCS-66.3
- [13] - Tubesheet MDMT, shell side, if applicable
- [14] - Tubesheet MDMT, tube side, if applicable
- [15] - Nozzle Material
- [16] - Shell or Head Material
- [17] - Impact Testing required
- [18] - Impact Testing not required, see UCS-66(b)(3)
- [20] - Cylinder/Cone Junction MDMT based on Longitudinal Stress considerations
- [21] - Body Flange Bolting Material
- [22] - Nozzle Flange Bolting Material
- [23] - Stiffening Ring to Shell Weld
- [24] - Saddle to Shell Weld

UG-84(b)(2) was not considered.

UCS-66(g) was not considered.

UCS-66(i) was not considered.

Notes:

- Impact test temps were not entered in and not considered in the analysis.
- UCS-66(i) applies to impact tested materials not by specification and
- UCS-66(g) applies to materials impact tested per UG-84.1 General Note (c).
- The Basic MDMT includes the (30F) PWHT credit if applicable.

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Class From To : Basic Element Checks.

Class From To: Check of Additional Element Data

Please insure the C factor entered is in accordance with Figure UG-34.

There were no geometry errors or warnings.

PV Elite performs all calculations internally in Imperial Units to remain compliant with the ASME Code and any built in assumptions in the ASME Code formulas. The finalized results are reflected to show the set of selected units for this analysis.

Information Regarding Nozzle Loads

Please note that nozzle loadings, if included, are assumed to be local in nature and will not contribute to or create a net section bending moment. Therefore, the addition of nozzle loads will not affect the support load calculation. If you wish to create a load on the support(s) from nozzle loads, you can enable that feature in tools->configuration->Job Specific Settings.



One or more nozzles in this model was specified using the actual thickness basis and a standard flange is also specified. In many cases a matching nominal diameter cannot be determined and the weight of the flange and optional blind flange cannot be determined. When actual thickness basis is specified, you must calculate the weight of the assembly (by hand) and enter it into the 'Overriding Weight' field in the nozzle dialog when there is a standard flange specified. However, for FVC nozzles that have their nominal diameter specified in the FVC dialog, entering the overriding weight for that nozzle is not required.



The level of precision is set to a value of less than 3. At this level you may not obtain proper resolution on some results. This value can be changed in Tools->Set Configuration Parameters->Default Values->Level of Precision.



The ASME Code Section VIII Division 1 does not explicitly require stresses on elements to be computed under test conditions. Please see interpretation VIII-1-10-12 for more information. PV Elite also computes weight due to test fluid and uses the weight in various calculations such as saddle and leg loads. PV Elite also computes static pressure due to operating and test liquids. The static pressure is used in conjunction with the design and test pressures to determine a total pressure in the determination of required thickness, MAWP etc. This ensures compliance with UG-22 requirements.

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Units used in this Analysis (SIASME):

Name	System Unit	Constant	User Unit
Length	Feet	304.8000	mm
Force	Pounds	4.4480	N
Mass	Pounds	0.4536	kgm
Area	sq. inches	645.1600	mm ²
Moment	ft. lbs.	1356.3000	N-mm
Stress	lbs./sq.in.	0.0069	MPa
Temperature	Degrees F	0.5556	°C
Pressure	psig	0.0069	MPa
Elast. Modulus	lbs./sq.in.	0.0069	MPa
Pipe Density	lbs./cu.in.	0.0277	kg/cm ³
Ins. Density	lbs./cu.ft.	0.1600E-04	kg/cm ³
Fluid Density	lbs./cu.ft.	0.1600E-04	kg/cm ³
Wind Speed	miles/hr	1.6093	km/hr
Tray Weight	lbs./sq.ft.	0.0005	kg/cm ²
Inertia	in.4	416231.0000	mm ⁴
G Load	G's	1.0000	g's
Wind Load	lbs./sq.ft.	0.0479	kPa
Elevation	Feet	304.8000	mm
Volume	in.3	0.0164	ltr
Diameter	inches	25.4000	mm
Thickness	inches	25.4000	mm

PV Elite Vessel Analysis Program: Input Data

Design Internal Pressure (for Hydrotest)	0.1	MPa
Design Internal Temperature	65.0	°C
Projection of Nozzle from Vessel Top	0	mm
Projection of Nozzle from Vessel Bottom	0	mm
Minimum Design Metal Temperature	11.4	°C
Type of Construction	Welded	
Special Service	None	
Degree of Radiography	RT-3	
Use Higher Longitudinal Stresses (Flag)	Y	
Select t for Internal Pressure (Flag)	N	
Select t for External Pressure (Flag)	N	
Select t for Axial Stress (Flag)	N	
Select Location for Stiff. Rings (Flag)	N	

User Spectrum Data points specified for increments of Time Period

Time Period	Sa/g
0.010	0.070
0.050	0.117
0.100	0.175
0.130	0.175
0.200	0.175
0.270	0.175
0.300	0.175
0.400	0.175
0.500	0.140
0.550	0.127
0.600	0.117
0.670	0.105
0.800	0.087
1.000	0.070
1.200	0.058
1.400	0.050
1.600	0.044
1.800	0.039
2.000	0.035
2.400	0.029
0.000	0.000



Consider Vortex Shedding

Y

Shop Pressure Test:

Type of Pressure Test User Entered Pressure
 Pressure Test Position Horizontal
 Test Performed in Corroded Condition No

Field Pressure Test:

Type of Pressure Test UG-99(b)
 Pressure Test Position Vertical
 Test Performed in Corroded Condition Yes

Load Case 1	NP+EW+WI+FW+BW
Load Case 2	NP+EW+EE+FS+BS
Load Case 3	NP+OW+WI+FW+BW
Load Case 4	NP+OW+EQ+FS+BS
Load Case 5	NP+HW+HI
Load Case 6	NP+HW+HE
Load Case 7	IP+OW+WI+FW+BW
Load Case 8	IP+OW+EQ+FS+BS
Load Case 9	EP+OW+WI+FW+BW
Load Case 10	EP+OW+EQ+FS+BS
Load Case 11	HP+HW+HI
Load Case 12	HP+HW+HE
Load Case 13	IP+WE+EW
Load Case 14	IP+WF+CW
Load Case 15	IP+VO+OW
Load Case 16	IP+VE+EW
Load Case 17	NP+VO+OW
Load Case 18	FS+BS+IP+OW
Load Case 19	FS+BS+EP+OW
Load Case 20	BL+IP+OW

Wind Design Code	IS 875 (PART 3):2015-2020	
Basic Wind Speed for IS-875	180	km/hr
Wind Zone Number	5	
Base Elevation	300	mm
Percent Wind for Hydrotest	33.0	
Risk Factor	[k1] 1.0	
Terrain Category	2	
Equipment Class	Class A	
Topography Factor	[k3] 1.0	
Importance Factor	[k4] 1.15	
Damping Factor (Beta) for Wind (Ope)	0.0100	
Damping Factor (Beta) for Wind (Empty)	0.0000	
Damping Factor (Beta) for Wind (Filled)	0.0000	

Seismic Design Code	IS 1893-RSM
Importance Factor for IS-1893	2.000
Soil Factor	1.000
Zone Number	2.000
Percent Seismic for Hydrotest	0.000
Period of Vibration (User defined) Sec	0.000
Damping Factor	2.000

User Spectrum Data points specified for increments of Time Period

Time Period	Sa/g
0.010	0.070
0.050	0.117
0.100	0.175
0.130	0.175
0.200	0.175
0.270	0.175
0.300	0.175
0.400	0.175
0.500	0.140
0.550	0.127
0.600	0.117

0.670	0.105
0.800	0.087
1.000	0.070
1.200	0.058
1.400	0.050
1.600	0.044
1.800	0.039
2.000	0.035
2.400	0.029
0.000	0.000

Consider Vertical Acceleration	Yes
Design Pressure + Static Head	Y
Consider MAP New and Cold in Noz. Design	N
Consider External Loads for Nozzle Des.	Y
Use ASME VIII-1 Appendix 1-9	N
Perform Blast Load Analysis	No
Material Database Year	Current w/Addenda or Code Year

Configuration Directives:

g0 is taken as min(g0, attached shell thk)	Yes
Do not use Nozzle MDMT Interpretation VIII-1 01-37	No
Use Table G instead of exact equation for "A"	Yes
Shell Head Joints are Tapered	Yes
Compute "K" in corroded condition	Yes
Use Code Case 2286	No
Use the MAWP to compute the MDMT	Yes
For thickness ratios <= 0.35, MDMT will be -155F (-104C)	Yes
For PWHT & P1 Materials the MDMT can be < -55F (-48C)	No
Using Metric Material Databases, ASME II D	Yes
Calculate B31.3 type stress for Nozzles with Loads	Yes
Reduce the MDMT due to lower membrane stress	Yes
Consider Longitudinal Stress in MDMT Calculations	Yes

Complete Listing of Vessel Elements and Details:

Element#	1/4
Element From Node	10
Element To Node	20
Element Type	Flat
Description	Bottom Head
Distance "FROM" to "TO"	18 mm
Element Outside Diameter	620 mm
Element Thickness	18 mm
Internal Corrosion Allowance	6 mm
Nominal Thickness	18 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	0.1 MPa
Design Temperature Internal Pressure	65 °C
Design External Pressure	0.0005 MPa
Design Temperature External Pressure	65 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70
Allowable Stress, Ambient	138 MPa
Allowable Stress, Operating	138 MPa
Allowable Stress, Hydrotest	234 MPa
Material Density	0.00775 kg/cm³
P Number Thickness	32.004 mm
Yield Stress, Operating	246 MPa
UCS-66 Chart Curve Designation	B
External Pressure Chart Name	CS-2
UNS Number	K02700
Product Form	Plate
Efficiency, Longitudinal Seam	0.85
Efficiency, Circumferential Seam	0.85
Flat Head Attachment Factor	0.3000001



Small diameter if Non-Circular 0 mm
 Weld is pre-Heated No

Element# 2/4
 Element From Node 20
 Element To Node 30
 Element Type Cylinder
 Description Shell-1
 Distance "FROM" to "TO" 657 mm
 Inside Diameter 600 mm
 Element Thickness 10 mm
 Internal Corrosion Allowance 6 mm
 Nominal Thickness 10 mm
 External Corrosion Allowance 0 mm
 Design Internal Pressure 0.1 MPa
 Design Temperature Internal Pressure 65 °C
 Design External Pressure 0.0005 MPa
 Design Temperature External Pressure 65 °C
 Effective Diameter Multiplier 1.2
 Material Name SA-516 70
 Efficiency, Longitudinal Seam 0.85
 Efficiency, Circumferential Seam 0.85
 Weld is pre-Heated No

Element From Node 20
 Detail Type Liquid
 Detail ID Liquid: 20
 Dist. from "FROM" Node / Offset dist 0 mm
 Height/Length of Liquid 500 mm
 Liquid Density 0.001861 kg/cm³

Element From Node 20
 Detail Type Nozzle
 Detail ID N14"
 Dist. from "FROM" Node / Offset dist 450 mm
 Nozzle Diameter 4 in.
 Nozzle Schedule 160
 Nozzle Class 300
 Layout Angle 270.0
 Blind Flange (Y/N) N
 Weight of Nozzle (Used if > 0) 0 N
 Grade of Attached Flange GR 1.1
 Nozzle Matl SA-106 B

Element From Node 20
 Detail Type Nozzle
 Detail ID N26"
 Dist. from "FROM" Node / Offset dist 450 mm
 Nozzle Diameter 6 in.
 Nozzle Schedule 120
 Nozzle Class 300
 Layout Angle 90.0
 Blind Flange (Y/N) N
 Weight of Nozzle (Used if > 0) 0 N
 Grade of Attached Flange GR 2.3
 Nozzle Matl SA-312 TP304L

Element From Node 20
 Detail Type Nozzle
 Detail ID N32"
 Dist. from "FROM" Node / Offset dist 200 mm
 Nozzle Diameter 2 in.
 Nozzle Schedule None
 Nozzle Class 300
 Layout Angle 90.0
 Blind Flange (Y/N) N
 Weight of Nozzle (Used if > 0) 106.72 N
 Grade of Attached Flange GR 2.3



Nozzle Matl	SA-106 B
Element From Node	20
Detail Type	Leg
Detail ID	LEGS
Dist. from "FROM" Node / Offset dist	100 mm
Diameter at Leg Centerline	689.55 mm
Leg Orientation	3
Number of Legs	3
Section Identifier	ISA75X75X8
Length of Legs	730 mm
Element From Node	20
Detail Type	Weight
Detail ID	Insert plate
Dist. from "FROM" Node / Offset dist	200 mm
Miscellaneous Weight	49.03 N
Offset from Element Centerline	0 mm
Element From Node	20
Detail Type	Weight
Detail ID	Inlet pipe N1
Dist. from "FROM" Node / Offset dist	450 mm
Miscellaneous Weight	294.18 N
Offset from Element Centerline	100 mm
Element From Node	20
Detail Type	Weight
Detail ID	Lifting Lug-1
Dist. from "FROM" Node / Offset dist	500 mm
Miscellaneous Weight	98.06 N
Offset from Element Centerline	300 mm
Element From Node	20
Detail Type	Weight
Detail ID	Lifting Lug-2
Dist. from "FROM" Node / Offset dist	500 mm
Miscellaneous Weight	98.06 N
Offset from Element Centerline	300 mm

Element#	3/4
Element From Node	30
Element To Node	40
Element Type	Flange
Description	Body Flange-1
Distance "FROM" to "TO"	90 mm
Flange Inside Diameter	600 mm
Element Thickness	75 mm
Internal Corrosion Allowance	6 mm
Nominal Thickness	75 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	0.1 MPa
Design Temperature Internal Pressure	65 °C
Design External Pressure	0.0005 MPa
Design Temperature External Pressure	65 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70
Perform Flange Stress Calculation (Y/N)	Y
Weld is pre-Heated	No

Element#	4/4
Element From Node	40
Element To Node	50
Element Type	Flange
Description	Body Flange-2
Distance "FROM" to "TO"	38 mm



Flange Outside Diameter	771	mm
Element Thickness	38	mm
Internal Corrosion Allowance	6	mm
Nominal Thickness	38	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	0.1	MPa
Design Temperature Internal Pressure	65	°C
Design External Pressure	0.0005	MPa
Design Temperature External Pressure	65	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	
Perform Flange Stress Calculation (Y/N)	Y	
Weld is pre-Heated	No	
Element From Node	40	
Detail Type	Nozzle	
Detail ID	N42"	
Dist. from "FROM" Node / Offset dist	0	mm
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	90.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	106.72	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	



XY Coordinate Calculations:

From	To	X (Horiz.) mm	Y (Vert.) mm	DX (Horiz.) mm	DY (Vert.) mm
Bottom Head		...	18	...	18
Shell-1		...	675	...	657
Body Flange-1		...	768.175	...	90
Body Flange-2		...	806.175	...	38

The calculated dimensions are based on the given element lengths. Due to variability in manufacturing (weld gaps etc.), the Tangent to Tangent length may not be exact. The length includes gasket thicknesses and raised face heights.

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Flange Input Data Values

Description: FLANGE-1 :

Body Flange-1

Flange Type	Integral Weld Neck		
Design Pressure	P	0.10	MPa
Design Temperature		65	°C
Internal Corrosion Allowance	ci	6.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Inside Diameter	B	600.000	mm
Flange Outside Diameter	A	771.000	mm
Flange Thickness	t	75.0000	mm
Thickness of Hub at Small End	go	10.0000	mm
Thickness of Hub at Large End	g1	13.0000	mm
Length of Hub	h	15.0000	mm
Flange Material		SA-516 70	
Flange Material UNS number		K02700	
Flange Allowable Stress At Temperature	Sfo	138.00	MPa
Flange Allowable Stress At Ambient	Sfa	138.00	MPa
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	172.00	MPa
Bolt Allowable Stress At Ambient	Sa	172.00	MPa
Diameter of Bolt Circle	C	712.000	mm
Nominal Bolt Diameter	a	24.0000	mm
Type of Threads		TEMA Metric	
Number of Bolts		24	
Flange Face Outside Diameter	Fod	663.000	mm
Flange Face Inside Diameter	Fid	600.000	mm
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	652.000	mm
Gasket Inside Diameter	Gi	626.000	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	68.95	MPa
Gasket Thickness	tg	3.1750	mm
Raised face depth	Fd	0.0000	mm

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Hub Small End Required Thickness due to Internal Pressure:

$$= (P \cdot (D/2 + Ca)) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (0.1 \cdot (600.0/2 + 6.0)) / (138.0 \cdot 1.0 - 0.6 \cdot 0.1) + Ca$$

$$= 6.2218 \text{ mm}$$

Small End Hub MAWP:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 4.0) / (306.0 + 0.6 \cdot 4.0)$$

$$= 1.790 \text{ MPa}$$

Hub Small End Hub MAPnc:

$$= (Sa \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 10.0) / (300.0 + 0.6 \cdot 10.0)$$

$$= 4.510 \text{ MPa}$$

Corroded Flange Thickness, tc	= T-ci	69.000	mm
Corroded Flange ID,	Bcor = B+2*ci	612.000	mm
Corroded Large Hub,	g1Cor = g1-ci	7.000	mm
Corroded Small Hub,	g0Cor = go-ci	4.000	mm
Code R Dimension,	R = ((C-Bcor)/2)-g1cor	43.000	mm
Gasket Contact Width,	N = (Go - Gi)/2	13.000	mm



Basic Gasket Width, $bo = N/2$ 6.500 mm
 Effective Gasket Width, $b = Cb \cdot \sqrt{bo}$ 6.425 mm
 Gasket Reaction Diameter, $G = Go - 2 \cdot b$ 639.151 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:
 $= \pi/4 \cdot G^2 \cdot P$
 $= \pi/4 \cdot 639.1509^2 \cdot 0.1$
 $= 32081.879 \text{ N}$

Contact Load on Gasket Surfaces [Hp or Wo,2]:
 $= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$
 $= 2 \cdot 6.4246 \cdot \pi \cdot 639.1509 \cdot 3.0 \cdot 0.1$
 $= 7739.471 \text{ N}$

Hydrostatic End Load at Flange ID [Hd]:
 $= \pi/4 \cdot Bcor^2 \cdot P$
 $= \pi/4 \cdot 612.0^2 \cdot 0.1$
 $= 29414.117 \text{ N}$

Pressure Force on Flange Face [Ht]:
 $= H - Hd$
 $= 32082 - 29414$
 $= 2667.762 \text{ N}$

Design Bolt Load for the Operating Condition [Wo]:
 $= \max(Wo,1 + Wo,2 + H'p, 0)$
 $= \max(32082 + 7739 + 0, 0)$
 $= 39821.348 \text{ N}$

Design Bolt Load for the Gasket Seating Condition [Wg]:
 $= y \cdot b \cdot \pi \cdot G + yPart \cdot bPart \cdot lp$
 $= 68.95 \cdot 6.4246 \cdot \pi \cdot 639.151 + 0.0 \cdot 0.0 \cdot 0.0$
 $= 889394.125 \text{ N}$

Required Bolt Area [Am]:
 $= \max((Wo + Fa + 4Me/G)/Sb, Wg/Sa)$
 $= \max((39821 + 0 + 4 \cdot 0/639.15)/172, 889394/172)$
 $= 5171.429 \text{ mm}^2$

ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmax]:
 $= 2a + 6t/(m + 0.5)$
 $= 2 \cdot 24.0 + 6 \cdot 69.0/(3.0 + 0.5)$
 $= 166.286 \text{ mm}$

Actual Circumferential Bolt Spacing [Bs]:
 $= C \cdot \sin(\pi / n)$
 $= 712.0 \cdot \sin(\pi/24)$
 $= 92.935 \text{ mm}$

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:
 $= \max(\sqrt{ Bs/(2a + t) }, 1)$
 $= \max(\sqrt{ 92.935/(2 \cdot 24.0 + 69.0) }, 1)$
 $= 1.0000$

Bolting Information for TEMA Metric Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	5171.429	7505.952	
Radial Distance between Hub and Bolts:	33.500	43.000	
Radial Distance between Bolts and Edge:	29.000	29.500	
Circ. Spacing between the Bolts:	57.500	92.935	166.286

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:
 $= Ab \cdot Sa/(y \cdot \pi(Go + Gi))$
 $= 7505.952 \cdot 172.0/(68.95 \cdot \pi(652.0 + 626.0))$
 $= 4.663 \text{ mm}$

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= Sa(Am + Ab) / 2$$

$$= 172.0(5171.4292 + 7505.9517) / 2$$

$$= 1090142.38 \text{ N}$$

Gasket Load for the Operating Condition [HG]:

$$= Wo - H$$

$$= 39821 - 32082$$

$$= 7739.47 \text{ N}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (712.0 - 639.1509) / 2$$

$$= 36.4245 \text{ mm}$$

Distance to Face Pressure Reaction [ht]:

$$= 0.5((C - B) / 2 + hg)$$

$$= 0.5((712.0 - 612.0) / 2 + 36.4245)$$

$$= 43.2123 \text{ mm}$$

Distance to End Pressure Reaction [hd]:

$$= (C - B - g1) / 2$$

$$= (712.0 - 612.0 - 7.0) / 2$$

$$= 46.5000 \text{ mm}$$

Summary of Moments for Internal Pressure: (N-mm)

Loading	Force	Distance	Bolt Corr	Moment
End Pressure, Md	29414.	46.5000	1.0000	1368311.
Face Pressure, Mt	2668.	43.2123	1.0000	115327.
Gasket Load, Mg	7739.	36.4245	1.0000	282021.
Gasket Seating, Matm	1090142.	36.4245	1.0000	39724040.

Total Moment for Operation, Mo 1765659. N-mm
 Total Moment for Gasket seating, Mg 39724040. N-mm

Effective Hub Length, $h_o = \sqrt{B_{cor} \cdot g_{oCor}}$ 49.477 mm
 Hub Ratio, $h/h_o = HL / H_o$ 0.303
 Thickness Ratio, $g1/g_o = (g1_{cor}/g_{oCor})$ 1.750

Flange Factors for Integral Flange:

Factor F per 4.16.5 0.874
 Factor V per 4.16.5 0.330
 Factor f per 4.16.5 1.584

Stress factor from Table 4.16.4 [K]:

$$= A / B$$

$$= 771.0 / 612.0$$

$$= 1.260$$

Stress factor from Table 4.16.4 [Y]:

$$= (0.66845 + (5.7169(K^2 \cdot \log_{10}(K))) / (K^2 - 1)) / (K - 1)$$

$$= (0.66845 + (5.7169(1.26^2 \cdot \log_{10}(1.26))) / (1.26^2 - 1)) / (1.26 - 1)$$

$$= 8.539$$

Stress factor from Table 4.16.4 [T]:

$$= (K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / ((1.0472 + 1.9448 \cdot K^2) \cdot (K - 1))$$

$$= (1.26^2(1 + 8.55246 \cdot \log_{10}(1.26)) - 1) / ((1.0472 + 1.9448 \cdot 1.26^2) \cdot (1.26 - 1))$$

$$= 1.814$$

Stress factor from Table 4.16.4 [U]:

$$= (K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / (1.36136(K^2 - 1) \cdot (K - 1))$$

$$= (1.26^2(1 + 8.55246 \cdot \log_{10}(1.26)) - 1) / (1.36136(1.26^2 - 1) \cdot (1.26 - 1))$$

$$= 9.384$$

Stress factor from Table 4.16.4 [Z]:

$$= (K^2 + 1) / (K^2 - 1)$$

$$= (1.26^2 + 1) / (1.26^2 - 1)$$

$$= 4.407$$

Stress factor from Table 4.16.4 for integral flanges [d]:

$$= U \cdot g_0^2 \cdot h_0 / V$$

$$= 9.384 \cdot 4.0^2 \cdot 49.477 / 0.33$$

$$= 22540.891 \text{ mm}^3$$

Stress factor from Table 4.16.4 for integral flanges [e]:

$$= F / h_0$$

$$= 0.874 / 49.477$$

$$= 0.018 \text{ 1/mm}$$

Stress factor from Table 4.16.4 [L]:

$$= (t \cdot e + 1) / T + t^3/d$$

$$= (69.0 \cdot 0.018 + 1) / 1.814 + 69.0^3 / 22540.891$$

$$= 15.797$$

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

$$= (f \cdot Mop / Bcor) / (L \cdot g_1^2)$$

$$= (1.5836 \cdot 1765659 / 612.0) / (15.797 \cdot 7.0^2)$$

$$= 5.90 \text{ MPa}$$

Longitudinal Hub Stress, Seating [SHa]:

$$= (f \cdot Matm / Bcor) / (L \cdot g_1^2)$$

$$= (1.5836 \cdot 39724040 / 612.0) / (15.797 \cdot 7.0^2)$$

$$= 132.75 \text{ MPa}$$

Radial Flange Stress, Operating [SRo]:

$$= (\beta \cdot Mop / Bcor) / (L \cdot t^2)$$

$$= (2.6256 \cdot 1765659 / 612.0) / (15.797 \cdot 69.0^2)$$

$$= 0.10 \text{ MPa}$$

Radial Flange Stress, Seating [SRa]:

$$= (\beta \cdot Matm / Bcor) / (L \cdot t^2)$$

$$= (2.6256 \cdot 39724040 / 612.0) / (15.797 \cdot 69.0^2)$$

$$= 2.27 \text{ MPa}$$

Tangential Flange Stress, Operating [STo]:

$$= (Y \cdot Mo / (t^2 \cdot Bcor)) - Z \cdot SRo$$

$$= (8.5394 \cdot 1765659 / (69.0^2 \cdot 612.0)) - 4.4065 \cdot 0.1$$

$$= 4.73 \text{ MPa}$$

Tangential Flange Stress, Seating [STa]:

$$= (y \cdot Matm / (t^2 \cdot Bcor)) - Z \cdot SRa$$

$$= (8.5394 \cdot 39724040 / (69.0^2 \cdot 612.0)) - 4.4065 \cdot 2$$

$$= 106.40 \text{ MPa}$$

Average Flange Stress, Operating [SAo]:

$$= (SHo + \max(SRo, STo)) / 2$$

$$= (6 + \max(0.1, 5)) / 2$$

$$= 5.31 \text{ MPa}$$

Average Flange Stress, Seating [SAa]:

$$= (SHa + \max(SRa, STa)) / 2$$

$$= (133 + \max(2, 106)) / 2$$

$$= 119.58 \text{ MPa}$$

Bolt Stress, Operating [BSO]:

$$= Wm1 / Ab$$

$$= 39821 / 7505.9517$$

$$= 5.31 \text{ MPa}$$

Bolt Stress, Seating [BSa]:

$$= (Wm2 / Ab)$$

$$= (889394 / 7505.9517)$$

$$= 118.50 \text{ MPa}$$

Flange Stress Analysis Results: MPa

	Actual	Operating Allowed	Gasket Seating Actual	Allowed
Longitudinal Hub	5.90	207.00	132.75	207.00
Radial Flange	0.10	138.00	2.27	138.00
Tangential Flange	4.73	138.00	106.40	138.00
Maximum Average	5.31	138.00	119.58	138.00
Bolting	5.31	172.00	118.50	172.00

Minimum Required Flange Thickness [Rigidity] 72.390 mm

Estimated M.A.W.P. (Operating) 2 MPa
 Estimated Finished Weight of Flange at given Thk. 109.6 kgm
 Estimated Unfinished Weight of Forging at given Thk 128.4 kgm

Flange rigidity based on required thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
 $= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 39724040 / 1.0 \cdot 1.0 \cdot 0.33 / (14.168 \cdot 202713 \cdot 4.0^2 \cdot 49.477 \cdot 0.3)$
 $= 1.000$ (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 $= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 1765659 / 1.0 \cdot 1.0 \cdot 0.33 / (14.168 \cdot 200199 \cdot 4.0^2 \cdot 49.477 \cdot 0.3)$
 $= 0.045$ (should be ≤ 1)

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
 $= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 39724040 / 1.0 \cdot 1.0 \cdot 0.33 / (15.797 \cdot 202713 \cdot 4.0^2 \cdot 49.477 \cdot 0.3)$
 $= 0.897$ (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 $= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 1765659 / 1.0 \cdot 1.0 \cdot 0.33 / (15.797 \cdot 200199 \cdot 4.0^2 \cdot 49.477 \cdot 0.3)$
 $= 0.040$ (should be ≤ 1)

Minimum Design Metal Temperature Results:

Thickness Ratio = 0.044, Temperature Reduction per Fig. UCS 66.1 = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Note: UCS-66(b)-(c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: MPa

	Actual	Operating Allowed	Gasket Seating Actual	Allowed
Longitudinal Hub	5.46	207.00	116.62	207.00
Radial Flange	0.31	138.00	6.59	138.00
Tangential Flange	3.09	138.00	65.96	138.00
Maximum Average	4.27	138.00	91.29	138.00
Bolting	5.31	172.00	118.50	172.00

Minimum Req'd Thickness, ambient, no Corrosion: 59.639 mm

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Flange Input Data Values Description: Flange-2 :

Body Flange-2

Flange Type		Blind	
Design Pressure	P	0.10	MPa
Design Temperature		65	°C
Internal Corrosion Allowance	ci	6.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Outside Diameter	A	771.000	mm
Flange Thickness	t	38.0000	mm
Flange Material		SA-516 70	
Flange Material UNS number		K02700	
Flange Allowable Stress At Temperature	Sfo	138.00	MPa
Flange Allowable Stress At Ambient	Sfa	138.00	MPa
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	172.00	MPa
Bolt Allowable Stress At Ambient	Sa	172.00	MPa
Perimeter along the Center of the Bolts	L	2236.814	mm
Diameter of Bolt Circle	C	712.000	mm
Nominal Bolt Diameter	a	24.0000	mm
Type of Threads		TEMA Metric	
Number of Bolts		24	
Flange Face Outside Diameter	Fod	663.000	mm
Flange Face Inside Diameter	Fid	600.000	mm
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	652.000	mm
Gasket Inside Diameter	Gi	626.000	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	68.95	MPa
Gasket Thickness	tg	3.1750	mm
Raised face depth	Fd	0.0000	mm

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Gasket Contact Width,	$N = (Go - Gi)/2$	13.000	mm
Basic Gasket Width,	$bo = N/2$	6.500	mm
Effective Gasket Width,	$b = Cb \cdot \sqrt{bo}$	6.425	mm
Gasket Reaction Diameter,	$G = Go - 2 \cdot b$	639.151	mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:
 $= \pi/4 \cdot G^2 \cdot P$
 $= \pi/4 \cdot 639.1509^2 \cdot 0.1$
 $= 32081.879 \text{ N}$

Contact Load on Gasket Surfaces [Hp or Wo,2]:
 $= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$
 $= 2 \cdot 6.4246 \cdot \pi \cdot 639.1509 \cdot 3.0 \cdot 0.1$
 $= 7739.471 \text{ N}$

Design Bolt Load for the Operating Condition [Wo]:
 $= \max(Wo,1 + Wo,2 + H'p, 0)$
 $= \max(32082 + 7739 + 0, 0)$
 $= 39821.348 \text{ N}$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$\begin{aligned}
 &= y \cdot b \cdot \pi \cdot G + y_{\text{Part}} \cdot b_{\text{Part}} \cdot l_p \\
 &= 68.95 \cdot 6.4246 \cdot \pi \cdot 639.151 + 0.0 \cdot 0.0 \cdot 0.0 \\
 &= 889394.125 \text{ N}
 \end{aligned}$$

Required Bolt Area [Am]:

$$\begin{aligned}
 &= \max((W_o + F_a + 4M_e/G) / S_b, W_g / S_a) \\
 &= \max((39821 + 0 + 4 \cdot 0 / 639.15) / 172, 889394 / 172) \\
 &= 5171.429 \text{ mm}^2
 \end{aligned}$$

Bolting Information for TEMA Metric Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	5171.429	7505.952	
Radial Distance between Bolts and Edge:	29.000	29.500	
Circ. Spacing between the Bolts:	57.500	92.935	...

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$\begin{aligned}
 &= A_b \cdot S_a / (y \cdot \pi (G_o + G_i)) \\
 &= 7505.952 \cdot 172.0 / (68.95 \cdot \pi (652.0 + 626.0)) \\
 &= 4.663 \text{ mm}
 \end{aligned}$$

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$\begin{aligned}
 &= S_a (A_m + A_b) / 2 \\
 &= 172.0 (5171.4292 + 7505.9517) / 2 \\
 &= 1090142.38 \text{ N}
 \end{aligned}$$

Gasket Load for the Operating Condition [HG]:

$$\begin{aligned}
 &= W_o \\
 &= 39821.35 \text{ N}
 \end{aligned}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$\begin{aligned}
 &= (C - G) / 2 \\
 &= (712.0 - 639.1509) / 2 \\
 &= 36.4245 \text{ mm}
 \end{aligned}$$

Flange Stresses

Tangential Flange Stress, Flat Head (4.6.2.3), Operating [STo]:

$$\begin{aligned}
 &= 1.9 \cdot W_{m1} \cdot hG \cdot B_{sc} / (t^2 \cdot G) + C \cdot Z \cdot P_{eq} \cdot G^2 / t^2 \\
 &= 1.9 \cdot 39821 \cdot 36.4245 \cdot 1.0 / (32.0^2 \cdot 639.1509) + 0.3 \cdot 1.0 \cdot 0.1 \cdot 639.1509^2 / 32.0^2 \\
 &= 16.18 \text{ MPa}
 \end{aligned}$$

Tangential Flange Stress, Flat Head (4.6.2.4), Seating [STa]:

$$\begin{aligned}
 &= 1.9 \cdot W \cdot hG \cdot B_{sc} / (t^2 \cdot G) \\
 &= 1.9 \cdot 1090142 \cdot 36.4245 \cdot 1.0 / (32.0^2 \cdot 639.1509) \\
 &= 115.28 \text{ MPa}
 \end{aligned}$$

Bolt Stress, Operating [BSo]:

$$\begin{aligned}
 &= W_{m1} / A_b \\
 &= 39821 / 7505.9517 \\
 &= 5.31 \text{ MPa}
 \end{aligned}$$

Bolt Stress, Seating [BSa]:

$$\begin{aligned}
 &= (W_{m2} / A_b) \\
 &= (889394 / 7505.9517) \\
 &= 118.50 \text{ MPa}
 \end{aligned}$$

Flange Stress Analysis Results: MPa

	Actual	Operating Allowed	Gasket Seating Actual	Gasket Seating Allowed
Tangential Flange	16.18	138.00	115.28	138.00
Bolting	5.31	172.00	118.50	172.00

Reqd. Blind Flange Thickness at Center 35.248 mm



Reqd. Blind Flange Thickness at Gasket 35.248 mm

Estimated M.A.W.P. (Operating) 1 MPa
 Estimated Finished Weight of Flange at given Thk. 137.5 kgm
 Estimated Unfinished Weight of Forging at given Thk 137.5 kgm

Minimum Design Metal Temperature Results:

Thickness Ratio = 0.117, Temperature Reduction per Fig. UCS 66.1 = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: MPa

	Operating		Gasket Seating	
	Actual	Allowed	Actual	Allowed
Tangential Flange	11.47	138.00	81.75	138.00
Bolting	5.31	172.00	118.50	172.00

Minimum Req'd Thickness, ambient, no Corrosion: 29.261 mm

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Internal Pressure Calculation Results:

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Internal Pressure Results for: Bottom Head

Flat Head Design Information:					
Design Pressure	P	0.109 MPa	Int. Design Temperature		65 °C
Material		SA-516 70	Ext. Design Temperature		65 °C
Length		18.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	6.0000 mm	Allowable Stress (ope)	S	138.0 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	138.0 MPa
Circular Diameter		620.000 mm	Actual Stress		100.2 MPa
Spec. Min. Thk t _e , or s _a		18.0000 mm	Spec. Nominal Thk.		18.000 mm
Surface Area		0.0 mm ²	SG of Contents		0.000
Element Volume		0.0 ltr	Weight of Contents		0.0 N
Empty Weight		413.0 N	Operating Weight		413.0 N
			Comp. All. Ope. Stress		142.0 MPa
			Attachment Factor	C	0.30

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	85 %	Long. Joint Eff.	E or v 85 %

Required Thickness due to Internal Pressure [tr]:

$$= d \cdot \sqrt{(Z \cdot C \cdot P) / (S \cdot E)} \text{ per UG-34 (c)(3)}$$

$$= 612.0 \cdot \sqrt{(1.0 \cdot 0.3 \cdot 0.109) / (138.0 \cdot 0.85)}$$

$$= 10.2247 + 6.0000 = 16.2247 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.009 MPa

$$= (t/d)^2 \cdot ((S \cdot E) / (C \cdot Z)) \text{ UG-34 (c)(3)}$$

$$= (12.0/612.0)^2 \cdot ((138.0 \cdot 0.85) / (0.3 \cdot 1.0))$$

$$= 0 - 0.01 = 0 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (t/d)^2 \cdot ((S \cdot E) / (C \cdot Z)) \text{ per UG-34 (c)(3)}$$

$$= (18.0/600.0)^2 \cdot ((138.0 \cdot 0.85) / (0.3 \cdot 1.0))$$

$$= 0 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (Z \cdot C \cdot P) / (((t/d)^2) \cdot E)$$

$$= (1.0 \cdot 0.3 \cdot 0.109) / (((12.0/612.0)^2) \cdot 0.85)$$

$$= 100.187 \text{ MPa}$$

Minimum Design Metal Temperature Results:

tg = 10.0, tgsr = 18.0, tr = 12.0, c = 6.0 mm, E* = 0.85
 Thickness Ratio = tr * E*/(tgsr - c) = 0.85, Temp. Reduction = 8 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -37 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	0.150 MPa
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	11 °C
Governing Thk.	10.000 mm	Basic MDMT	-29 °C
Stress/Thk. Ratio	0.850	Temperature Difference	8 °C
UCS-66(c) red.	No	Computed Min. Temp.	-37 °C

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MDMT per UG-20(f)	-29 °C
Joint Efficiency E*	85 %

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Internal Pressure Calculation Results:

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Internal Pressure Results for: Shell-1

Cylindrical Shell Design Information:					
Design Pressure	P	0.109 MPa	Int. Design Temperature		65 °C
Material		SA-516 70	Ext. Design Temperature		65 °C
Length		657.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	6.0000 mm	Allowable Stress (ope)	S	138.0 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	138.0 MPa
Inside Diameter	Di	600.000 mm	Actual Stress		9.9 MPa
Spec. Min. Thk t,e, or sa		10.0000 mm	Spec. Nominal Thk.		10.000 mm
Surface Area		1279696.4 mm²	SG of Contents		1.864
Element Volume		185.8 ltr	Weight of Contents		2582.9 N
Empty Weight		956.9 N	Operating Weight		3539.8 N
			Comp. All. Ope. Stress		118.4 MPa

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	85 %	Long. Joint Eff.	E or v 85 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (0.109 \cdot 306.0) / (138.0 \cdot 0.85 - 0.6 \cdot 0.109)$$

$$= 0.2849 + 6.0000 = 6.2849 \text{ mm}$$

Note: The thickness required was less than the Code Minimum, therefore the Code Minimum value of 1.5000 mm per UG-16 will be used.

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.009 MPa

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 0.85 \cdot 4.0) / (306.0 + 0.6 \cdot 4.0)$$

$$= 2 - 0.01 = 2 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 0.85 \cdot 10.0) / (300.0 + 0.6 \cdot 10.0)$$

$$= 4 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (0.109 \cdot (306.0 + 0.6 \cdot 4.0)) / (0.85 \cdot 4.0)$$

$$= 9.899 \text{ MPa}$$

% Elongation per Table UG-79-1 (50*tnom/Rf*(1-Rf/Ro)) 1.639 %

Minimum Design Metal Temperature Results:

Govrn. thk, tg = 10.0, tr = 1.5, c = 6.0 mm, E* = 0.85

Thickness Ratio = tr * E*/(tg - c) = 0.319, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	0.150 MPa



UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	11 °C
Governing Thk.	10.000 mm	Basic MDMT	-29 °C
Stress/Thk. Ratio	0.319	Temperature Difference	75 °C
UCS-66(c) red.	No	Computed Min. Temp.	-104 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	85 %

EXTERNAL PRESSURE RESULTS for: Shell-1

Cylindrical Shell From 20 to 30 Ext. Chart: CS-2 at 65 °C

Elastic Modulus from Chart: CS-2 at 65 °C: 199955 MPa

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
4.000	620.00	657.00	155.00	1.0597	0.0006542	65.41

$$MAEP = (4*B)/(3*(Do/t)) = (4*65.4077)/(3*155.0) = 0.5626 \text{ MPa}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
0.243	620.00	657.00	2554.50	1.0597	0.0000096	0.96

$$MAEP = (4*B)/(3*(Do/t)) = (4*0.958)/(3*2554.499) = 0.0005 \text{ MPa}$$

Results for Maximum Stiffened Length					Slen	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
4.000	620.00	22038.68	155.00	35.5463	0.0000464	4.64

$$MAEP = (4*B)/(3*(Do/t)) = (4*4.6418)/(3*155.0) = 0.0399 \text{ MPa}$$

Hydrostatic Test Pressure Results:

UG99(b) = $1.30 * \text{M.A.W.P.} * \text{Sa/S}$ 0.184 MPa
 UG99b(b)(35) = $1.30 * \text{Design Pres} * \text{Sa/S}$ 0.130 MPa
 UG99(c) = $1.30 * \text{M.A.P.} - \text{Head(Hyd)}$ 0.450 MPa
 UG100 = $1.10 * \text{M.A.W.P.} * \text{Sa/S}$ 0.155 MPa
 PED = $\max(1.43*DP, 1.25*DP*ratio)$ 0.143 MPa
 PED&UG99(b)(35) = $\max(1.43*DP, 1.3*DP*ratio)$ 0.143 MPa
 App 27-4 = M.A.W.P. 0.141 MPa

User Defined Hydrostatic Test Pressure at High Point 0.140 MPa

Horizontal Test performed per: User Hydro Pressure

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (MPa):

From To	Stress	Allowable	Ratio	Pressure
Bottom Head	59.6	234.0	0.255	0.15
Shell-1	5.3	234.0	0.022	0.15

Stress ratios for Nozzle and Pad Materials (MPa):

Description	Pad/Nozzle	Ambient	Operating	Ratio
N1_4"	Nozzle	118.00	118.00	1.000
N2_6"	Nozzle	115.00	105.00	1.095
N3_2"	Nozzle	118.00	118.00	1.000
N3_2"	Pad	138.00	138.00	1.000
N4_2"	Nozzle	118.00	118.00	1.000
N4_2"	Pad	138.00	138.00	1.000

Minimum 1.000

Stress ratios for Pressurized Vessel Elements (MPa):

Description	Ambient	Operating	Ratio
Bottom Head	138.00	138.00	1.000
Shell-1	138.00	138.00	1.000
Body Flange-1	138.00	138.00	1.000
Body Flange-2	138.00	138.00	1.000

Minimum 1.000

Hoop Stress in Nozzle Wall during Pressure Test (MPa):

Description	Ambient	Operating	Ratio
N1_4"	0.56	216.00	0.003
N2_6"	0.80	153.00	0.005
N3_2"	0.36	216.00	0.002
N4_2"	0.37	216.00	0.002

Elements Suitable for Test Pressure.

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Hydrostatic Test Pressure Results:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	0.184 MPa
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	0.130 MPa
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	0.450 MPa
UG100	= 1.10 * M.A.W.P. * Sa/S	0.155 MPa
PED	= max(1.43*DP, 1.25*DP*ratio)	0.143 MPa
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	0.143 MPa
App 27-4	= M.A.W.P.	0.141 MPa

UG-99(b), Test Pressure Calculation:
 = Test Factor • MAWP • Stress Ratio
 = 1.3 • 0.141 • 1.0
 = 0.184 MPa

Vertical Test performed per: UG-99b

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (MPa):

From To	Stress	Allowable	Ratio	Pressure
Bottom Head	175.7	234.0	0.751	0.19
Shell-1	17.3	234.0	0.074	0.19

Stress ratios for Nozzle and Pad Materials (MPa):

Description	Pad/Nozzle	Ambient	Operating	Ratio
N1_4"	Nozzle	118.00	118.00	1.000
N2_6"	Nozzle	115.00	105.00	1.095
N3_2"	Nozzle	118.00	118.00	1.000
N3_2"	Pad	138.00	138.00	1.000
N4_2"	Nozzle	118.00	118.00	1.000
N4_2"	Pad	138.00	138.00	1.000

Minimum 1.000

Stress ratios for Pressurized Vessel Elements (MPa):

Description	Ambient	Operating	Ratio
Bottom Head	138.00	138.00	1.000
Shell-1	138.00	138.00	1.000
Body Flange-1	138.00	138.00	1.000
Body Flange-2	138.00	138.00	1.000

Minimum 1.000

Hoop Stress in Nozzle Wall during Pressure Test (MPa):

Description	Ambient	Operating	Ratio
N1_4"	0.73	216.00	0.003
N2_6"	1.05	153.00	0.007
N3_2"	0.48	216.00	0.002
N4_2"	0.46	216.00	0.002

Elements Suitable for Test Pressure.

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External Pressure Calculation Results Summary:

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External Pressure Calculations:

From	To	Section Length mm	Outside Diameter mm	Corroded Thickness mm	Factor A	Factor B MPa
10	20	No Calc	...	12	No Calc	No Calc
20	30	657	620	4	0.00065422	65.4077
30	40	No Calc	...	69	No Calc	No Calc
40	50	No Calc	...	32	No Calc	No Calc

External Pressure Calculations:

From	To	External Actual T. mm	External Required T. mm	External Design Pressure MPa	External M.A.W.P. MPa
10	20	18	No Calc	0.00050003	No Calc
20	30	10	6.24271	0.00050003	0.56265
30	40	75	72.39	0.00050003	No Calc
40	50	38	35.2552	0.00050003	No Calc

Minimum

1

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners mm	Allowable Length Bet. Stiffeners mm	Ring Inertia Required mm^4	Ring Inertia Available mm^4
10	20	No Calc	No Calc	No Calc	No Calc
20	30	657	22038.7	No Calc	No Calc
30	40	No Calc	No Calc	No Calc	No Calc
40	50	No Calc	No Calc	No Calc	No Calc

Elements Suitable for External Pressure.

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Element and Detail Weights:

From	To	Element Metal Wgt. kgm	Element ID Volume ltr	Corroded Metal Wgt. kgm	Corroded ID Volume ltr	Extra due Misc % kgm
10	20	42.1188	...	28.0792	...	2.94831
20	30	97.5831	185.796	39.417	193.302	6.83081
30	40	109.601	26.3083	101.633	26.4797	7.67205
40	50	137.503	...	137.503	...	9.62521
Total		386	212.10	306	219.78	27

Weight of Details:

From	Type	Weight of Detail kgm	X Offset, Dtl. Cent. mm	Y Offset, Dtl. Cent. mm	Z Offset, Dtl. Cent. mm	Description
20	Liqd	263.404	...	250	...	Liquid: 20
20	Nozl	18.2313	-357.836	450	...	N1_4"
20	Nozl	31.5335	356.312	450	...	N2_6"
20	Nozl	11.6449	414.719	200	...	N3_2"
20	Legs	35.7554	...	-265	...	LEGS
20	Wght	5	...	200	...	Insert plate
20	Wght	30	100	450	...	Inlet pipe N1
20	Wght	10	...	500	-300	Lifting Lug-1
20	Wght	10	...	500	300	Lifting Lug-2
40	Nozl	11.6449	...	152.719	...	N4_2"

Total Weight of Each Detail Type:

Liquid	263.4
Nozzles	73.1
Legs	35.8
Weights	55.0
Sum of the Detail Weights	427.2 kgm

Weight Summation Results: (kgm)

	Fabricated	Shop Test	Shipping	Erected	Empty	Operating
Main Elements	413.9	413.9	413.9	413.9	413.9	413.9
Nozzles	73.1	73.1	73.1	73.1	73.1	73.1
Legs	35.8	35.8	35.8	35.8	35.8	35.8
Wld Weights	55.0	55.0	55.0	55.0	55.0	55.0
Ope. Liquid	...	...	...	...	...	263.4
Test Liquid	...	212.0	...	...	...	...
Totals	577.7	789.7	577.7	577.7	577.7	841.1

Miscellaneous Weight Percent: 7.0 %

Note that the above value for the miscellaneous weight percent has been applied to the shells/heads/flange/tubesheets/tubes etc. in the weight calculations for metallic components.

Weight Summary:

Fabricated Wt.	- Bare weight without removable internals	577.7 kgm
Shop Test Wt.	- Fabricated weight + water (full)	789.7 kgm
Shipping Wt.	- Fab. weight + removable intls.+ shipping app.	577.7 kgm
Erected Wt.	- Fab. wt + or - loose items (trays,platforms etc.)	577.7 kgm
Ope. Wt. no Liq	- Fab. weight + internals. + details + weights	577.7 kgm
Operating Wt.	- Empty weight + operating liq. uncorroded	841.1 kgm
Field Test Wt.	- Empty weight + test medium (full)	789.7 kgm

Weights used in Vortex Shedding Calculations:

Mass of the Upper 1/3 of the Vertical Vessel (operating)	276.4 kgm
Mass of the Upper 1/3 of the Vertical Vessel (filled w/water)	342.8 kgm
Mass of the Upper 1/3 of the Vertical Vessel (empty)	297.9 kgm

Surface Areas of Elements:

From	To	Outside Surface Area mm ²	Inside Surface Area mm ²
10	20	...	...
20	30	1279696	1238415
30	40	395292	169646
40	50	558915	...

Total	2233903.2	1408061.8 mm ²
Total	2.2	1.4 m ²

Element and Detail Weights:

From	To	Total Ele. Empty Wgt. kgm	Total. Ele. Oper. Wgt. kgm	Total. Ele. Hydro. Wgt. kgm	Total Dtl. Offset Mom. N-mm	Oper. Wgt. No Liquid kgm
10	20	45.0671	45.0671	45.0671	...	45.0671
20	Legs	33.6109	73.7028	61.873	93600.2	33.6109
Legs	30	187.213	410.524	344.633	...	187.213
30	40	117.273	117.273	143.565	...	117.273
40	50	158.773	158.773	158.773	...	158.773

Cumulative Vessel Weight

From	To	Cumulative Ope Wgt. No Liquid kgm	Cumulative Oper. Wgt. kgm	Cumulative Hydro. Wgt. kgm
10	20	...	...	...
20	Legs	-45.0671	-45.0671	-45.0671
Legs	30	463.259	686.57	646.971
30	40	276.046	276.046	302.338
40	50	158.773	158.773	158.773

The cumulative operating weights no liquid in the column above are the cumulative operating weights minus the operating liquid weight minus any weights absent in the empty condition.

Cumulative Vessel Moment

From	To	Cumulative Empty Mom. N-mm	Cumulative Oper. Mom. N-mm	Cumulative Hydro. Mom. N-mm
10	20	...	...	...
20	Legs	93600.2	93600.2	93600.2
Legs	30	...	...	...
30	40	...	...	...
40	50	...	...	...

Nozzle Flange MAWP Results: (MPa & °C)

Nozzle Description	Flange Rating		Des. Temp	Class	Grade/Group	Equiv. Press	Max Pressure		
	Ope.	Ambient					UG-44	50%	DNV
N1_4"	4.89	5.10	65	300	GR 1.1	4.040	3.291	4.887	3.291
N2_6"	3.89	4.11	65	300	GR 2.3	3.171	2.657	3.885	2.657
N3_2"	3.89	4.11	65	300	GR 2.3	5.668	0.160	3.885	0.160
N4_2"	4.89	5.10	65	300	GR 1.1	...	...	...	...
Min Rating	0.160	4.110, [for Core Elements]					...	...	...

Selected Method for Derating ASME B16.5 Flange MAWP: ASME UG-44(b)

The UG-44(b) Method is based on the paper PVP 2013-97814. PV Elite uses the maximum loads from each load category to determine ME and FE. In many cases, the computed maximum allowable pressure will be greater than the flange rating. In these cases, the minimum of the rating from the table and the UG-44(b) method will be used. SA-193 B8 Cl. 2 bolts or ones with higher allowable stresses at the specified bolt size shall be used.

The axial force FE must put the nozzle in tension (-P for 107/297; +Fr for PD 5500 AnnexG) to be considered.

Nozzle Description	Gasket Dimension G mm	Fm Value	Resolved Moment ME N-mm	Axial Force FE N	MAWP PD MPa	Maximum Pressure Pmax MPa	Pass Fail
N1_4"	138.176	0.50	2000000	2700	0.141	3.291	Pass
N2_6"	196.088	0.50	4500001	4000	0.141	2.657	Pass
N3_2"	77.851	0.50	500000	1300	0.141	0.160	Pass

Evaluating Flange on Nozzle [N14"]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot ((PR - PD) + FM \cdot PR)$$

$$= 16(2000000) + 4 \cdot \text{abs}(2700)138.176 \leq \pi \cdot 138.176^3 \cdot ((4.887 - 0.141) + .500 \cdot 4.887)$$

$$= 0.33493E+08 \leq 0.59608E+08 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle N14" [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(4.887, 0.500 \cdot 4.887 + 4.887 - 33492926.0 / (\pi \cdot 138.176^3))$$

$$= \min(4.887, 3.291)$$

$$= 3.291 \text{ MPa}$$

Evaluating Flange on Nozzle [N26"]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot ((PR - PD) + FM \cdot PR)$$

$$= 16(4500001) + 4 \cdot \text{abs}(4000)196.088 \leq \pi \cdot 196.088^3 \cdot ((3.885 - 0.141) + .500 \cdot 3.885)$$

$$= 0.75139E+08 \leq 0.13474E+09 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle N26" [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(3.885, 0.500 \cdot 3.885 + 3.885 - 75138680.0 / (\pi \cdot 196.088^3))$$

$$= \min(3.885, 2.657)$$

$$= 2.657 \text{ MPa}$$

Evaluating Flange on Nozzle [N32"]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot ((PR - PD) + FM \cdot PR)$$

$$= 16(500000) + 4 \cdot \text{abs}(1300)77.851 \leq \pi \cdot 77.851^3 \cdot ((3.885 - 0.141) + .500 \cdot 3.885)$$

$$= 0.84050E+07 \leq 0.84324E+07 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle N32" [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(3.885, 0.500 \cdot 3.885 + 3.885 - 8404991.0 / (\pi \cdot 77.851^3))$$

$$= \min(3.885, 0.160)$$

$$= 0.160 \text{ MPa}$$

Flange Pressure Ratings are per ASME B16.5 2025 Metric Edition

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The Natural Frequencies for the vessel have been computed iteratively by solving a system of matrices. These matrices describe the mass and the stiffness of the vessel. This is the generalized eigenvalue/eigenvector problem and is referenced in some mathematical texts.

The Natural Frequency for the Vessel (Empty.) is 48.9616 Hz.

The Natural Frequency for the Vessel (Ope...) is 42.2632 Hz.

The Natural Frequency for the Vessel (Filled) is 42.8887 Hz.

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Wind Load Calculation Results:

Wind Load Calculations per India Std. IS 875 (PART 3):2015, Amd. 1&2 (2020):

Input Values:

Wind Design Code	IS 875 (PART 3):2015-2020	
Basic Wind Speed for IS-875	180	km/hr
Wind Zone Number	5	
Base Elevation	300	mm
Percent Wind for Hydrotest	33.0	
Risk Factor	[k1]	1.0
Terrain Category		2
Equipment Class		Class A
Topography Factor	[k3]	1.0
Importance Factor	[k4]	1.15
Damping Factor (Beta) for Wind (Ope)		0.0100
Damping Factor (Beta) for Wind (Empty)		0.0000
Damping Factor (Beta) for Wind (Filled)		0.0000
Use Gust Response Factor (Dynamic Analysis)		No

Calculated Values:

Vessel Height to Diameter ratio: 1.295

Checking the requirement for dynamic effect of wind (9.1):

Vessel Operating Natural Frequency: 42.263 Hz

Note: Vessel Natural frequency is ≥ 1 Hz.

Note: Vessel height to diameter ratio (1.295) is ≤ 5 .

Note: Dynamic wind analysis is not required.

Design Wind Speed [Vz]:

$$= \text{Basic Wind Speed} \cdot k_1 \cdot k_2 \cdot k_3 \cdot k_4 \quad (\text{for the 1st element})$$

$$= 50.0 \cdot 1.0 \cdot 1.0 \cdot 1.0 \cdot 1.15$$

$$= 57.496 \text{ meters/s}$$

Design Wind Pressure per 4.5.2 [pd]:

$$= \max(0.7 \cdot p_z, K_d \cdot K_a \cdot K_c \cdot 0.6 \cdot V_z^2)$$

$$= \max(0.7 \cdot 1983.503, 1.0 \cdot 1.0 \cdot 1.0 \cdot 0.6 \cdot 57.5^2)$$

$$= 1983.5 \text{ N/m}^2$$

Element Wind Load (operating condition):

$$= \text{Wind Area} \cdot C_f \cdot p_d$$

$$= 0.0 \cdot 0.7 \cdot 1983.503$$

$$= 18.6 \text{ N}$$

From	Height mm	k1	k2	k3	Vz m/sec	Cf
10	309.00	1.0000	1.0000	1.0000	57.50	0.7000
20	646.50	1.0000	1.0000	1.0000	57.50	0.7000
30	1023.17	1.0000	1.0000	1.0000	57.50	0.7000
40	1087.18	1.0000	1.0000	1.0000	57.50	0.7000

Wind Vibration Calculations

This evaluation is based on work by Kanti Mahajan and Ed Zorilla

Nomenclature

- Cf - Correction factor for natural frequency
- D - Average outer insulated diameter of vessel, mm
- Df - Damping Factor < 0.75 Unstable, > 0.95 Stable
- Dr - Average outer insulated of top half of vessel, mm



f - Natural frequency of vibration (Hertz)
 f1 - Natural frequency of bare vessel based on a unit value of $(D/L^2)(10^4)$
 L - Total height of structure mm
 Lc - Total length of conical section(s) of vessel, mm
 tb - Uncorroded plate thickness at bottom of vessel, mm
 V30 - Design Wind Speed provided by user, km/hr
 Vc - Critical wind velocity, km/hr
 Vw - Maximum wind speed at top of structure, km/hr
 W - Total corroded weight of structure, N
 Ws - Cor. vessel weight excl. weight of parts which do not effect stiff. N
 Z - Maximum amplitude of vibration at top of vessel, mm
 D1 - Logarithmic decrement (taken as 0.03 for Welded Structures)
 Vp - Vib. Chance, ≤ 0.000003 (High); $0.000003 < 0.000004$ (Probable)
 P30 - wind pressure 30 feet above the base

Check other Conditions and Basic Assumptions:

:: Total Cone Length / Total Length < 0.5

:: $0.0/806.175 = 0.0$

:: $[D / L^2] \cdot 10^4 < 8$ (English Units, ft.)

:: $[2.04/2.64^2] \cdot 10^4 = 2919.146$ [Geometry Violation]

Compute the vibration possibility. If $V_p > 0.000004$ no chance. [Vp]:

$= W / (L \cdot Dr^2)$

$= 7407 / (806.18 \cdot 670.826^2)$

$= 0.20416E-04$

Since V_p is > 0.000004 no further vibration analysis is required !

Wind Loads on Masses/Equipment/Piping

ID	Wind Area mm ²	Elevation mm	Pressure kPa	Force N
Insert plate	0.00	518.00	1.98	0.00
Inlet pipe N1	0.00	768.00	1.98	0.00
Lifting Lug-1	0.00	818.00	1.98	0.00
Lifting Lug-2	0.00	818.00	1.98	0.00

The Natural Frequency for the Vessel (Ope...) is 42.2632 Hz.

Wind Load Calculation:

From	To	Wind Height mm	Wind Diameter mm	Wind Area mm ²	Wind Pressure kPa	Element Wind Load N
10	20	309	744	13392	1.98426	20.4536
20	30	646.5	744	488808	1.98426	746.555
30	40	1023.17	720	64800	1.98426	98.9688
40	50	1087.18	925.2	35157.6	1.98426	53.6961

The Wind Loads calculated and printed in the Wind Load calculation report have been factored by the input scalar/load reduction factor of: 1.100.

Be sure the wind speed is in accordance with the specified wind design code.

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Earthquake Load Calculation Results:

Seismic Analysis Results per IS 1893 (Part 4): 2024

Input Data:

Importance Factor as Entered by User I 2.0000
 Zone Number 2
 Soil Type Rocky or Hard Soil
 Response reduction factor (Force factor) R 2.0000
 Time Period of Vibration (program or user) 0.0300 Seconds
 Value of Sa/g from user Spectrum Data 0.0935
 Percent Damping 2.0 %
 Considering MCE for load determination No

Calculated results:

Horizontal Seismic Coefficient per 7.3.1 [Ah]:
 $= (Sa/g)_{ss} \cdot DampVal \cdot (I/R)$
 $= (0.0935 \cdot 1.0) \cdot 1.0$
 $= 0.093$

User Spectrum Data points specified for increments of Time Period

Time Period	Sa/g
0.010	0.070
0.050	0.117
0.100	0.175
0.130	0.175
0.200	0.175
0.270	0.175
0.300	0.175
0.400	0.175
0.500	0.140
0.550	0.127
0.600	0.117
0.670	0.105
0.800	0.087
1.000	0.070
1.200	0.058
1.400	0.050
1.600	0.044
1.800	0.039
2.000	0.035
2.400	0.029
0.000	0.000

Actual Time Period = 0.03 secs
 Computed Sa/g = 0.0935

Note:

It was indicated to ignore the damping factor, assuming that the Site Spectrum specified is per the actual damping value of this vessel.
 The value of "DampVal" has been set to 1 (not used).

Earthquake Element Load, for the first Element:
 $= \text{Earthquake Weight} \cdot Ah \cdot \text{Scalar}$
 $= 441.93 \cdot 0.0935 \cdot 1.0$
 $= 41.3 \text{ N}$

Vertical load per 8.4:
 $= 2/3 \cdot Ah \cdot W \cdot \text{Scalar}$
 $= 2/3 \cdot 0.0935 \cdot 7897 \cdot 1.0$
 $= 492.3 \text{ N}$

The Natural Frequency for the Vessel (Ope...) is 42.2632 Hz.

Earthquake Load Calculation:



From	To	Earthquake Height mm	Earthquake Weight N	Element Ope Load N	Element Emp Load N	Element Vertical Load N
10	20	9	441.928	41.3202	41.3202	-27.5468
20	Legs	118	722.729	67.5752	30.8165	-72.5969
Legs	30	396.5	4025.6	376.394	171.648	419.66
30	40	723.175	1149.98	107.523	107.523	168.73
40	50	787.175	1556.93	145.573	145.573	97.0486

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The following table is for the Operating Case.

Wind/Earthquake Shear, Bending:

From	To	Distance to Support mm	Cumulative Wind Shear N	Earthquake Shear N	Wind Bending N-mm	Earthquake Bending N-mm
10	20	109	...	...	...	...
20	Legs	50	20.4536	41.3202	184.157	372.033
Legs	30	278.5	919.673	738.385	264291	259532
30	40	605.175	152.665	253.096	10310.7	20714.4
40	50	669.175	53.6961	145.573	1020.64	2767.01

Note:

The Wind Shears/Moments and the Earthquake Shears/Moments calculated and printed in the Wind/Earthquake Shear and Bending report have been factored by the input Scalar/Load reductions factors of;
 Wind: 1.100; Earthquake: 1.000.

Note:

Review the Vessel Design Summary for the cumulative shear force and bending moment on the support.

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Wind Deflection Calculations:

The following table is for Wind Deflection; Operating Case.

Wind Load Deflection:

From	To	Cumulative Shear N	Centroid Deflection mm	Elem. End Deflection mm	Elem. Ang. Rotation
10	20	...	0.33126	0.33126	0.0006803
20	Legs	20.4536	0.33126	0.33128	0.0006807
Legs	30	919.673	0.33151	0.33187	0.00068164
30	40	152.665	0.33193	0.33199	0.00068164
40	50	53.6961	0.33201	0.33204	0.00068164

Critical Wind Velocity for Tower Vibration:

From	To	1st Crit. Wind Speed km/hr	2nd Crit. Wind Speed km/hr
10	20	564.464	3527.9
20	30	564.464	3527.9
30	40	546.255	3414.09
40	50	701.938	4387.11

Allowable deflection at the Tower Top (Ope)(6.000"/100ft. Criteria)
 Allowable deflection : 4.031 Actual deflection : 0.332 mm

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Longitudinal Stress Constants:

From	To	Metal Area New mm ²	Metal Area Corroded mm ²	Section Modulus New mm ³	Section Modulus Corroded mm ³
10	20	36078	24278.2	5601004	3838623
20	30	19163.7	7740.87	2876103	1184452
30	40	19163.7	7740.87	2876103	1184452
40	50	19163.7	7740.87	2876103	1184452

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Longitudinal Allowable Stresses:

From	To	Tensile MPa	Hydrotest Tensile MPa	Compressive MPa	Hydrotest Compressive MPa
10	20	140.757	238.676	-141.97	-141.97
20	Legs	140.757	238.676	-118.367	-118.367
Legs	30	140.757	238.676	-118.367	-118.367
30	40	165.597	280.795	-118.367	-118.367
40	50	165.597	280.795	-118.367	-118.367

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Longitudinal Stress Report



Longitudinal Operating and Empty Stresses are computed in the corroded condition. Stresses due to loads in the hydrostatic test cases have also been computed in the corroded condition.

Longitudinal Pressure Stresses due to:

From	To	Longitudinal Stress Internal Pressure MPa	Longitudinal Stress External Pressure MPa	Longitudinal Stress Hydrotest Pressure MPa
10	20	1.29667	-0.0069611	2.37996
20	30	3.805	-0.019502	6.98387
30	40	...	...	...
40	50	...	...	...

Longitudinal Stresses due to Weight Loads for these Conditions:

From	To	Wght. Str. Empty MPa	Wght. Str. Operating MPa	Wght. Str. Hydrotest MPa	Wght. Str. Emp. Mom. MPa	Wght. Str. Opr. Mom. MPa
10	20	...	...	...	...	...
20	Legs	0.057095	0.057095	0.057095	0.078999	0.078999
Legs	30	-0.5869	-0.53611	-0.5869	...	...
30	40	-0.34972	-0.34972	-0.34972	...	...
40	50	-0.20115	-0.20115	-0.20115	...	...

Longitudinal Stresses due to Weight Loads and Bending for these Conditions:

From	To	Wght. Str. Hyd. Mom. MPa	Bend. Str. Oper. Wind MPa	Bend. Str. Oper. Equ. MPa	Bend. Str. Hyd. Wind MPa	Bend. Str. Hyd. Equ. MPa
10	20	...	...	...	...	...
20	Legs	0.078999	0.00015543	0.000314	0.51291E-04	...
Legs	30	...	0.22306	0.21905	0.07361	...
30	40	...	0.0087022	0.017483	0.0028717	...
40	50	...	0.00086142	0.0023354	0.00028427	...

Longitudinal Stresses due to these Conditions:

From	To	Vortex Shedding Operating Case MPa	Vortex Shedding Empty Case MPa	Vortex Shedding Test Case MPa	Earthquake Empty Case MPa
10	20	...	...	...	...
20	Legs	...	...	...	0.000314
Legs	30	...	...	...	0.17245
30	40	...	...	...	0.017483
40	50	...	...	...	0.0023354

Longitudinal Stresses due to Applied Axial Forces:

From	To	Longitudinal Stress Y Forces Wind MPa	Longitudinal Stress Y Forces Seismic MPa
10	20	...	-0.0011347
20	Legs	...	-0.0093792
Legs	30	...	0.054218
30	40	...	0.021799

40	50	...	0.012538
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Longitudinal Stresses due to User Forces and Moments:

From	To	Wind For/Mom Corroded MPa	Earthquake For/Mom Corroded MPa	Wind For/Mom No Corrosion MPa	Earthquake For/Mom No Corrosion MPa
10	20	...	...	...	...
20	Legs	...	...	...	...
Legs	30	...	...	...	...
30	40	...	...	...	...
40	50	...	...	...	...

Bending Stresses due to Blast Load:

From	To	Bending Stress MPa
10	20	...
20	Legs	...
Legs	30	...
30	40	...
40	50	...

Stress Combination Load Cases for Vertical Vessels:

Load Case Definition Key

IP = Longitudinal Stress due to Internal Pressure
 EP = Longitudinal Stress due to External Pressure
 HP = Longitudinal Stress due to Hydrotest Pressure
 NP = No Pressure
 EW = Longitudinal Stress due to Weight (No Liquid)
 OW = Longitudinal Stress due to Weight (Operating)
 HW = Longitudinal Stress due to Weight (Hydrotest)
 WI = Bending Stress due to Wind Moment (Operating)
 EQ = Bending Stress due to Earthquake Moment (Operating)
 EE = Bending Stress due to Earthquake Moment (Empty)
 HI = Bending Stress due to Wind Moment (Hydrotest)
 HE = Bending Stress due to Earthquake Moment (Hydrotest)
 WE = Bending Stress due to Wind Moment (Empty) (no CA)
 WF = Bending Stress due to Wind Moment (Filled) (no CA)
 CW = Longitudinal Stress due to Weight (Empty) (no CA)
 VO = Bending Stress due to Vortex Shedding Loads (Ope)
 VE = Bending Stress due to Vortex Shedding Loads (Emp)
 VF = Bending Stress due to Vortex Shedding Loads (Test No CA.)
 FW = Axial Stress due to Vertical Forces for the Wind Case
 FS = Axial Stress due to Vertical Forces for the Seismic Case
 BW = Bending Stress due to Lat. Forces for the Wind Case, Corroded
 BS = Bending Stress due to Lat. Forces for the Seismic Case, Corroded
 BN = Bending Stress due to Lat. Forces for the Wind Case, UnCorroded
 BU = Bending Stress due to Lat. Forces for the Seismic Case, UnCorroded
 BL = Bending Stress due to Blast Loads, Corroded

General Notes:

Case types HI and HE are in the Corroded condition.

Case types WE, WF, and CW are in the Un-Corroded condition.

A blank stress and stress ratio indicates that the corresponding stress comprising those components that did not contribute to that type of stress.

An asterisk (*) in the final column denotes overstress.

Analysis of Load Case 1 : NP+EW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	0.00	140.76	0.00	141.97	0.0000	-0.0000
20	0.14	140.76	-0.02	118.37	0.0010	0.0002
20		140.76	-0.81	118.37		0.0068

Analysis of Load Case 2 : NP+EW+EE+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		140.76	-0.00	141.97		0.0000
20	0.13	140.76	-0.03	118.37	0.0009	0.0003
20		140.76	-0.71	118.37		0.0060

Analysis of Load Case 3 : NP+OW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	0.00	140.76	0.00	141.97	0.0000	-0.0000
20	0.14	140.76	-0.02	118.37	0.0010	0.0002
20		140.76	-0.76	118.37		0.0064

Analysis of Load Case 4 : NP+OW+EQ+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
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10		140.76	-0.00	141.97		0.0000
20	0.13	140.76	-0.03	118.37	0.0009	0.0003
20		140.76	-0.70	118.37		0.0059

Analysis of Load Case 5 : NP+HW+HI

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	0.00	238.68	0.00	141.97	0.0000	-0.0000
20	0.14	238.68	-0.02	118.37	0.0006	0.0002
20		238.68	-0.66	118.37		0.0056

Analysis of Load Case 6 : NP+HW+HE

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	0.00	238.68	0.00	141.97	0.0000	-0.0000
20	0.14	238.68	-0.02	118.37	0.0006	0.0002
20		238.68	-0.59	118.37		0.0050

Analysis of Load Case 7 : IP+OW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	1.30	140.76		141.97	0.0092	
20	3.94	140.76		118.37	0.0280	
20	3.49	140.76		118.37	0.0248	

Analysis of Load Case 8 : IP+OW+EQ+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	1.30	140.76		141.97	0.0092	
20	3.93	140.76		118.37	0.0279	
20	3.54	140.76		118.37	0.0252	

Analysis of Load Case 9 : EP+OW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		140.76	-0.01	141.97		0.0000
20	0.12	140.76	-0.04	118.37	0.0008	0.0004
20		140.76	-0.78	118.37		0.0066

Analysis of Load Case 10 : EP+OW+EQ+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		140.76	-0.01	141.97		0.0001
20	0.11	140.76	-0.05	118.37	0.0008	0.0004
20		140.76	-0.72	118.37		0.0061

Analysis of Load Case 11 : HP+HW+HI

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	2.38	238.68		141.97	0.0100	
20	7.12	238.68		118.37	0.0298	
20	6.47	238.68		118.37	0.0271	

Analysis of Load Case 12 : HP+HW+HE

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	2.38	238.68		141.97	0.0100	
20	7.12	238.68		118.37	0.0298	
20	6.40	238.68		118.37	0.0268	

Analysis of Load Case 13 : IP+WE+EW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
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Node	Stress	Stress	Stress	Stress	Ratio	Ratio
10	1.30	140.76		141.97	0.0092	
20	3.94	140.76		118.37	0.0280	
20	3.22	140.76		118.37	0.0229	

Analysis of Load Case 14 : IP+WF+CW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	1.30	140.76		141.97	0.0092	
20	3.83	140.76		118.37	0.0272	
20	3.59	140.76		118.37	0.0255	

Analysis of Load Case 15 : IP+VO+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	1.30	140.76		141.97	0.0092	
20	3.94	140.76		118.37	0.0280	
20	3.27	140.76		118.37	0.0232	

Analysis of Load Case 16 : IP+VE+EW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	1.30	140.76		141.97	0.0092	
20	3.94	140.76		118.37	0.0280	
20	3.22	140.76		118.37	0.0229	

Analysis of Load Case 17 : NP+VO+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	0.00	140.76	0.00	141.97	0.0000	-0.0000
20	0.14	140.76	-0.02	118.37	0.0010	0.0002
20		140.76	-0.54	118.37		0.0045

Analysis of Load Case 18 : FS+BS+IP+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	1.30	117.30		118.31	0.0110	
20	3.93	117.30		98.64	0.0335	
20	3.32	117.30		98.64	0.0283	

Analysis of Load Case 19 : FS+BS+EP+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		117.30	-0.01	118.31		0.0001
20	0.11	117.30	-0.05	98.64	0.0009	0.0005
20		117.30	-0.50	98.64		0.0051

Analysis of Load Case 20 : BL+IP+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	1.30	140.76		141.97	0.0092	
20	3.94	140.76		118.37	0.0280	
20	3.27	140.76		118.37	0.0232	

Absolute Maximum of the all of the Stress Ratio's 0.0335

Governing Element: Shell-1

Governing Load Case 18 : FS+BS+IP+OW

Element Description	Long Pressure Stress	Weight + Axial Load Stress	Eccentric Moment Stress	Wind or Earthquake Stress	Combined Stress	Which Side
---------------------	----------------------	----------------------------	-------------------------	---------------------------	-----------------	------------



Bottom Head	1.3	...	...	...	1.3	+ Tens
Bottom Head	1.3	...	...	...	...	- Comp
Shell-1	3.8	...	0.1	...	3.9	+ Tens
Shell-1	3.8	...	-0.1	...	...	- Comp
Shell-1	3.8	-0.5	...	...	3.3	+ Tens
Shell-1	3.8	-0.5	...	...	...	- Comp



Shop/Field Installation Options :

Note : The CG is computed from the first Element From Node

Center of Gravity of the Liquid	268.000 mm
Center of Gravity of the Nozzles	500.341 mm
Center of Gravity of the Legs	-247.000 mm
Center of Gravity of the Added Weights (Operating)	463.455 mm
Center of Gravity of the Added Weights (Empty)	463.455 mm
Center of Gravity of Bare Shell New and Cold	573.133 mm
Center of Gravity of Bare Shell Corroded	638.055 mm
Vessel CG in the Operating Condition	441.072 mm
Vessel CG in the Fabricated (Shop/Empty) Condition	502.725 mm
Vessel CG in the Test Condition	473.330 mm

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RESULTS FOR LEGS : Operating Case Description: LEGS

Legs attached to: Shell-1

Section Properties : Single Angle ISA75X75X8

India ISI Structural Steel Data

Overall Leg Length		730.000	mm
Effective Leg Length	Leglen	500.000	mm
Distance Leg Up Side of Vessel		100.000	mm
Number of Legs	Nleg	3	
Cross Sectional Area for ISA75X75X8	Aleg	1137.998	mm ²
Section Inertia (strong axis)		589997.000	mm ⁴
Section Inertia (weak axis)		589997.000	mm ⁴
Section Modulus (strong axis)		11000.045	mm ³
Section Modulus (weak axis)		11000.045	mm ³
Radius of Gyration (strong axis)		22.800	mm
Radius of Gyration (weak axis)		22.800	mm

Leg Orientation - Diagonal

Overturning Moment at top of Legs		293720.9	N-mm
Total Weight Load at top of Legs	W	7920.1	N
Total Shear force at top of Legs		919.7	N
Additional force in Leg due to Bracing	Fadd	0.0	N
Occasional Load Factor	Occfac	1.333	
Effective Leg End Condition Factor	k	1.000	

Note: The Legs are Not Cross Braced

The Leg Shear Force includes Wind, Seismic, and Applied Load Effects.

Pad Width along Circumference	C11P	120.000	mm
Pad Length along Vessel Axis	C22P	120.000	mm
Pad Thickness	Tpad	10.000	mm

Maximum Shear at top of one Leg [Vleg]:
 $= (\max(\text{Wind, Seismic, Blast}) + \text{applied forces}) (I_{\max} / I_{\text{tot}})$
 $= (919.7) (957450 / 1450184)$
 $= 607.19 \text{ N}$

Axial Compression, Leg furthest from the Neutral Axis [Sma]:
 $= (W / N_{\text{leg}} + (M_{\text{leg}} / (N_{\text{legm}} \cdot R_n))) / A_{\text{leg}}$
 $= (7920 / 3 + (293604 / (1 \cdot 344.776))) / 1137.998$
 $= 3.07 \text{ MPa}$

Axial Compression, Leg closest to the Neutral Axis [Sva]:
 $= (W / N_{\text{leg}}) / A_{\text{leg}}$
 $= (7920 / 3) / 1137.998$
 $= 2.32 \text{ MPa}$

Computing Principal Axis and Inertias for Angle:

Leg lengths and thickness:	75.000	75.000	8.0000
Distance to geometric centroid:	21.400	21.400	
Arm about YY:	17.400	20.100	
Arm about ZZ:	16.100	17.400	
Leg areas:	600.00	536.00	
Geometric inertia components YY:	0.18485E+06	0.41705E+06	
Geometric inertia components ZZ:	0.43677E+06	0.16514E+06	
Geometric inertias Iy & Iz:	0.60191E+06	0.60191E+06	
Product of inertia:	0.35554E+06		
Mohrs Radius:	0.35554E+06		
Average Inertia:	0.60191E+06		

QFACT = 1.0000 FBZ = 0.16383
 Principal Axis Inertias (Z&W) = 0.24637E+06 0.95745E+06
 Angle to Principal Axis = 45.000



Distances to extreme fibers CW & CZ = 53.033 22.769
 FOB from Eq 5-5 = 3.1165
 Bending allowables Fby & Fbz = 0.16383 0.16383

Shear Center Coordinates Wo & Zo: 23.978 0.0000

Values for Elastic Flexural-Torsional Buckling Stress:

E, G, J, R0²: 29500. 11346. 0.58326E-01 2.5308
 AREA, LENGTH, Kw, Kz: 1.7639 19.685 1.0000 1.0000
 H, Few, Fez, Fej: 0.64788 979.84 252.13 148.24
 Fe computed from C4-1: 140.02

Initial (Kl/r)max, & (Kl/r)equiv = 33.982 45.599
 Final (Kl/r)max, & Cc = 45.599 127.18
 Fa based on Eq 4-1 = 0.12937

Leg Bending and Axial Stresses vs. Allowables: (MPa)

	Actual	Allowable
Weak Axis Bending :	19.84	218.38
Strong Axis Bending :	11.89	218.38
Axial Compression :	3.07	172.45

UNITY CHECKS ARE: H1-1 0.000
 H1-2 0.000
 H1-3 0.163

AISC Unity Check : 0.163, should be <= to 1 for Code compliance, [Passed]

Bolting Size Requirement for Leg Baseplates :

Baseplate Material	SA-516 70
Baseplate Allowable Stress	SBA 137.90 MPa
Baseplate Length	B 200.0000 mm
Baseplate Width	D 200.0000 mm
Baseplate Thickness	BTHK 15.0000 mm
Leg Dimension Along Baseplate Length	d 74.9998 mm
Leg Dimension Along Baseplate Width	b 74.9998 mm
Bolt Circle Diameter	BCD 440.0000 mm
Bolt Material	SA-193 B7
Bolt Allowable Tensile Stress	STBA 137.30 MPa
Bolt Allowable Shear Stress	SBShear 107.87 MPa
Anchor Bolt Nominal Diameter	BOD 20.0000 mm
Number of Anchor Bolts in Tension per Leg	NB 0
Total Number of Anchors Bolt per Leg	NBT 1
Bolt Corrosion Allowance	bca 1.5000 mm
Ultimate 28-day Concrete Strength	FCPRIME 8.750 MPa

Leg Baseplate and Bolting Analysis, including Moments

Shear Stress in a Single Bolt [tb]:
 = Shear Force/(Bolt Area • Number of Bolts)
 = 919.7/(145.781 • 3)
 = 2.1 MPa. Must be less than 107.9 MPa.

Plate thickness required (1/2"; 13mm) per D. Moss

Since there is only 1 bolt, the standard computation for this geometry is not possible. A reasonable bolt size should be selected. Per Megyesy, use a 1 inch (25.4 mm) bolt.

Review notes about nozzle loadings and supports in the Warnings report.

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RESULTS FOR LEGS : HydroTest Case Description: LEGS

Legs attached to: Shell-1

Section Properties : Single Angle ISA75X75X8

India ISI Structural Steel Data

Overall Leg Length		730.000	mm
Effective Leg Length	Leglen	500.000	mm
Distance Leg Up Side of Vessel		100.000	mm
Number of Legs	Nleg	3	
Cross Sectional Area for ISA75X75X8	Aleg	1137.998	mm ²
Section Inertia (strong axis)		589997.000	mm ⁴
Section Inertia (weak axis)		589997.000	mm ⁴
Section Modulus (strong axis)		11000.045	mm ³
Section Modulus (weak axis)		11000.045	mm ³
Radius of Gyration (strong axis)		22.800	mm
Radius of Gyration (weak axis)		22.800	mm

Leg Orientation - Diagonal

Overturning Moment at top of Legs		116645.9	N-mm
Total Weight Load at top of Legs	W	7415.8	N
Total Shear force at top of Legs		303.5	N
Additional force in Leg due to Bracing	Fadd	0.0	N
Occasional Load Factor	Occfac	1.333	
Effective Leg End Condition Factor	k	1.000	

Note: The Legs are Not Cross Braced

The Leg Shear Force includes Wind, Seismic, and Applied Load Effects.

Pad Width along Circumference	C11P	120.000	mm
Pad Length along Vessel Axis	C22P	120.000	mm
Pad Thickness	Tpad	10.000	mm

Maximum Shear at top of one Leg [Vleg]:
 $= (\max(\text{Wind, Seismic, Blast}) + \text{applied forces}) (I_{\max} / I_{\text{tot}})$
 $= (303.5) (957450 / 1450184)$
 $= 200.37 \text{ N}$

Axial Compression, Leg furthest from the Neutral Axis [Sma]:
 $= (W / N_{\text{leg}} + (M_{\text{leg}} / (N_{\text{legm}} \cdot R_n))) / A_{\text{leg}}$
 $= (7416 / 3 + (116600 / (1 \cdot 344.776))) / 1137.998$
 $= 2.47 \text{ MPa}$

Axial Compression, Leg closest to the Neutral Axis [Sva]:
 $= (W / N_{\text{leg}}) / A_{\text{leg}}$
 $= (7416 / 3) / 1137.998$
 $= 2.17 \text{ MPa}$

Computing Principal Axis and Inertias for Angle:

Leg lengths and thickness:	75.000	75.000	8.0000
Distance to geometric centroid:	21.400	21.400	
Arm about YY:	17.400	20.100	
Arm about ZZ:	16.100	17.400	
Leg areas:	600.00	536.00	
Geometric inertia components YY:	0.18485E+06	0.41705E+06	
Geometric inertia components ZZ:	0.43677E+06	0.16514E+06	
Geometric inertias Iy & Iz:	0.60191E+06	0.60191E+06	
Product of inertia:	0.35554E+06		
Mohrs Radius:	0.35554E+06		
Average Inertia:	0.60191E+06		

QFACT = 1.0000 FBZ = 0.16383
 Principal Axis Inertias (Z&W) = 0.24637E+06 0.95745E+06
 Angle to Principal Axis = 45.000



Distances to extreme fibers CW & CZ = 53.033 22.769
 FOB from Eq 5-5 = 3.1165
 Bending allowables Fby & Fbz = 0.16383 0.16383

Shear Center Coordinates Wo & Zo: 23.978 0.0000

Values for Elastic Flexural-Torsional Buckling Stress:

E, G, J, R0²: 29500. 11346. 0.58326E-01 2.5308
 AREA, LENGTH, Kw, Kz: 1.7639 19.685 1.0000 1.0000
 H, Few, Fez, Fej: 0.64788 979.84 252.13 148.24
 Fe computed from C4-1: 140.02

Initial (Kl/r)max, & (Kl/r)equiv = 33.982 45.599
 Final (Kl/r)max, & Cc = 45.599 127.18
 Fa based on Eq 4-1 = 0.12937

Leg Bending and Axial Stresses vs. Allowables: (MPa)

	Actual	Allowable
Weak Axis Bending :	6.55	218.38
Strong Axis Bending :	3.92	218.38
Axial Compression :	2.47	172.45

UNITY CHECKS ARE: H1-1 0.000
 H1-2 0.000
 H1-3 0.062

AISC Unity Check : 0.062, should be <= to 1 for Code compliance, [Passed]

Bolting Size Requirement for Leg Baseplates :

Baseplate Material	SA-516 70
Baseplate Allowable Stress	SBA 137.90 MPa
Baseplate Length	B 200.0000 mm
Baseplate Width	D 200.0000 mm
Baseplate Thickness	BTHK 15.0000 mm
Leg Dimension Along Baseplate Length	d 74.9998 mm
Leg Dimension Along Baseplate Width	b 74.9998 mm
Bolt Circle Diameter	BCD 440.0000 mm
Bolt Material	SA-193 B7
Bolt Allowable Tensile Stress	STBA 137.30 MPa
Bolt Allowable Shear Stress	SBShear 107.87 MPa
Anchor Bolt Nominal Diameter	BOD 20.0000 mm
Number of Anchor Bolts in Tension per Leg	NB 0
Total Number of Anchors Bolt per Leg	NBT 1
Bolt Corrosion Allowance	bca 1.5000 mm
Ultimate 28-day Concrete Strength	FCPRIME 8.750 MPa

Leg Baseplate and Bolting Analysis, including Moments

Shear Stress in a Single Bolt [tb]:
 = Shear Force/(Bolt Area • Number of Bolts)
 = 919.7/(145.781 • 3)
 = 2.1 MPa. Must be less than 107.9 MPa.

Plate thickness required (1/2"; 13mm) per D. Moss

Since there is only 1 bolt, the standard computation for this geometry is not possible. A reasonable bolt size should be selected. Per Megyesy, use a 1 inch (25.4 mm) bolt.

Review notes about nozzle loadings and supports in the Warnings report.

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Input, Nozzle Desc: N14"
From: 20

Pressure for Reinforcement Calculations	P	0.1009	MPa
Temperature for Internal Pressure	Temp	65	°C
Design External Pressure	Pext	0.00	MPa
Temperature for External Pressure	Tempex	65	°C

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Cylindrical Shell	D	600.00	mm
Design Length of Section	L	657.0000	mm
Shell Finished (Minimum) Thickness	t	14.0000	mm
Shell Internal Corrosion Allowance	c	6.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		468.00	mm
-----------------------------------	--	--------	----

User Entered Minimum Design Metal Temperature		11.40	°C
---	--	-------	----

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		270.00	deg
Diameter		4.0000	in.

Size and Thickness Basis		Nominal	
Nominal Thickness		160	

Flange Material		SA-105	
Flange Type		Weld Neck Flange	

Corrosion Allowance	can	6.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	115.6720	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	10.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: N14"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation		3.438	in.
Actual Thickness Used in Calculation		0.531	in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 0.1 \cdot 306.0 / (138 \cdot 1.0 - 0.6 \cdot 0.1)$
 $= 0.2239$ mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 0.1 \cdot 49.66 / (118 \cdot 1.0 - 0.6 \cdot 0.1)$
 $= 0.0425 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.0439 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 198.6504 mm
 Parallel to Vessel Wall, opening length d 99.3252 mm
 Normal to Vessel Wall (Thickness Limit), no pad Tlnp 18.7185 mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118.0 / 138.0)$
 $= 0.855$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118.0 / 138.0)$
 $= 0.855$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.855, 1.0)$
 $= 0.855$

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	22.722	12.317	NA
Area in Shell	A1	755.490	753.659	NA
Area in Nozzle Wall	A2	238.321	238.276	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds A41+A42+A43		85.507	85.507	NA
Area in Element	A5	0.000	0.000	NA
TOTAL AREA AVAILABLE	Atot	1079.318	1077.442	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (99.3252 \cdot 0.2239 \cdot 1.0 + 2 \cdot 7.4874 \cdot 0.2239 \cdot 1.0 \cdot (1 - 0.86))$
 $= 22.722 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 99.325(1.0 \cdot 8.0 - 1.0 \cdot 0.224) - 2 \cdot 7.487$
 $(1.0 \cdot 8.0 - 1.0 \cdot 0.2239) \cdot (1 - 0.855)$
 $= 755.490 \text{ mm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp)(tn - trn)fr2$
 $= (2 \cdot 18.72)(7.49 - 0.04) \cdot 0.8551$
 $= 238.321 \text{ mm}^2$

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi - can / 0.707)^2 \cdot fr2$
 $= 10.0^2 \cdot 0.8551 + (0.0)^2 \cdot 0.8551$
 $= 85.507 \text{ mm}^2$



UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $t_a = 6.0439$ mm
 Wall Thickness per UG16(b), $tr_{16b} = 7.5000$ mm
 Wall Thickness, shell/head, internal pressure $trb_1 = 6.2239$ mm
 Wall Thickness $tb_1 = \max(trb_1, tr_{16b}) = 7.5000$ mm
 Wall Thickness $tb_2 = \max(trb_2, tr_{16b}) = 7.5000$ mm
 Wall Thickness per table UG-45 $tb_3 = 11.2578$ mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb_3, \max(tb_1, tb_2)]$
 $= \min[11.258, \max(7.5, 7.5)]$
 $= 7.5000$ mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(t_a, t_b)$
 $= \max(6.0439, 7.5)$
 $= 7.5000$ mm

Available Nozzle Neck Thickness = $0.875 * 13.487 = 11.801$ mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	31.0,	Allowable	:	118.0 MPa	Passed
Expansion	:	0.0,	Allowable	:	264.0 MPa	Passed
Occasional	:	0.3,	Allowable	:	156.9 MPa	Passed
Shear	:	22.1,	Allowable	:	82.6 MPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, $t_g = 11.801$, $tr = 0.042$, $c = 6.0$ mm, $E^* = 1.0$
 Thickness Ratio = $tr * E^* / (t_g - c) = 0.007$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-3 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, $t_g = 11.801$, $tr = 0.042$, $c = 6.0$ mm, $E^* = 1.0$
 Thickness Ratio = $tr * E^* / (t_g - c) = 0.007$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-24 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $0.10/5.10 = 0.020$

Weld Size Calculations, Description: N14"

Intermediate Calc. for nozzle/shell Welds $T_{min} = 7.4874$ mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$5.2412 = 0.7 * t_{min}$	$7.0700 = 0.7 * W_o$ mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$
 $= \max(0, (22.7215 - 755.4897 + 2 \cdot 7.4874 \cdot 0.8551 \cdot (1.0 \cdot 8.0 - 0.2239)) \cdot 138)$
 $= \max(0, -87372.38) \text{ N}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(W_i-\text{Can}/.707)^2 \cdot f_{r2}) \cdot S_v$
 $= (238.3215 + 0.0 + 85.5072 - 0.0 \cdot 0.86) \cdot 138$
 $= 44683.76 \text{ N}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$
 $= (238.3215 + 0.0 + 85.5072 + (102.4363)) \cdot 138$
 $= 58818.52 \text{ N}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$
 $= (238.3215 + 0.0 + 85.5072 + 0.0 + (102.4363)) \cdot 138$
 $= 58818.52 \text{ N}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.1416/2.0) \cdot 114.3 \cdot 10.0 \cdot 0.49 \cdot 118$
 $= 103800. \text{ N}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$
 $= (3.1416 \cdot 53.4063) \cdot (13.4874 - 6.0) \cdot 0.7 \cdot 118$
 $= 103755. \text{ N}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot D_{lo} \cdot (T - \text{cas}) \cdot 0.74 \cdot S_{ng}$
 $= (3.1416/2.0) \cdot 114.3 \cdot (14.0 - 6.0) \cdot 0.74 \cdot 138$
 $= 146664. \text{ N}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (\text{SONW} + \text{SNW}) = (103800 + 103755) = 207555 \text{ N}$

Failure Path [Path 2-2]:
 $= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$
 $= (103800 + 0 + 146664 + 0) = 250464 \text{ N}$

Failure Path [Path 3-3]:
 $= (\text{Sonw} + \text{Tngw} + \text{Sinw})$
 $= (103800 + 146664 + 0) = 250464 \text{ N}$

Summary of Failure Path Calculations:

Path 1-1 = 207555 N	, must exceed W = 0 N	or W1 = 44683 N
Path 2-2 = 250464 N	, must exceed W = 0 N	or W2 = 58818 N
Path 3-3 = 250464 N	, must exceed W = 0 N	or W3 = 58818 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 2 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0.001 MPa

The Drop for this Nozzle is : 5.4939 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 135.1658 mm

This nozzle or its diameter limit intersects weld seam # [3] on this element. The distance between the nozzle and seam is, 88.01 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 99.33 mm

Input Echo, WRC297 Item 1, Description: N14" :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	6.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	600.000	mm
Shell Thickness	14.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc	138.00 MPa
Vessel Hot S.I. Allowable	Smh	138.00 MPa

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	6.0000	mm
Nozzle Diameter	87.325	mm
Nozzle Thickness	13.4874	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-106 B	
Nozzle Cold S.I. Allowable	SNmc	118.00 MPa
Nozzle Hot S.I. Allowable	SNmh	118.00 MPa

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

*External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSained loads.*

Design Internal Pressure	Dp	0.10	MPa
Radial Load	P	-2700.00	N
Circumferential Shear	Vc	4000.00	N
Longitudinal Shear	Vl	4000.00	N
Circumferential Moment	Mc	0.120E+07	N-mm
Longitudinal Moment	Ml	0.160E+07	N-mm
Torsional Moment	Mt	0.200E+07	N-mm

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

*Warning - The ratio, Dn/Tn (8.475) is < 10
 check the Limitations of WRC bulletin 297.*

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$$

$$= 114.3 + 2 \cdot 1.65 \cdot \sqrt{(310.0(14.0 - 6.0))}$$

$$= 278.639 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	620.000 mm
Nozzle Outside Diameter	(d) =	114.300 mm
Vessel Thickness used	(T) =	8.000 mm
Nozzle Thickness used	(t) =	7.487 mm
T / t	=	1.068
d / t	=	15.266

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.623$$

$$Nr/P = 0.083$$



Mr/P = 0.135
 MØ/P = 0.048
 NØ/P = 0.117
 MrD/Mc = 0.257
 NrDL/Mc = 0.126
 MØd/Mc = 0.099
 NØDL/Mc = 0.074
 MrD/Ml = 0.141
 NrDL/Ml = 0.080
 MØD/Ml = 0.054
 NØDL/Ml = 0.159

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	B1 Bottom Inside
Outplane Membrane (P)	4	4	4	4
Outplane Bending (P)	12	-12	12	-12
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-34	-34	34	34
Outplane Bending (ML)	-70	70	70	-70
Normal Pressure Stress	3	3	3	3
Outplane Stress Summary	-84	32	125	-38

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	B1 Bottom Inside
Inplane Membrane (P)	3	3	3	3
Inplane Bending (P)	34	-34	34	-34
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-17	-17	17	17
Inplane Bending (ML)	-184	184	184	-184
Inplane Pressure Stress	1	1	1	1
Inplane Stress Summary	-162	138	241	-196

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	B1 Bottom Inside
Outplane Shear (Vc)	2	2	-2	-2
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	12	12	12	12
Shear Stress Summary	14	14	9	9

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	B1 Bottom Inside
Two * Max Shear Stress	165	140	242	196

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	D1 Right Inside
Outplane Membrane (P)	4	4	4	4
Outplane Bending (P)	12	-12	12	-12
Outplane Membrane (Mc)	-12	-12	12	12
Outplane Bending (Mc)	-97	97	97	-97
Outplane Membrane (ML)	0	0	0	0



Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	1	1	1	1

Outplane Stress Summary	-90	79	128	-90
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Inplane Membrane (P)	3	3	3	3
Inplane Bending (P)	34	-34	34	-34
Inplane Membrane (Mc)	-20	-20	20	20
Inplane Bending (Mc)	-252	252	252	-252
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	3	3	3	3

Inplane Stress Summary	-231	205	314	-259
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-2	-2	2	2
Torsional Shear (Mt)	12	12	12	12

Shear Stress Summary	9	9	14	14
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Two * Max Shear Stress	232	206	316	260
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Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
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Hoop Membrane (P)	4	4	4	4
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-34	-34	34	34
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	0	0	0	0

Hoop Stress Summary	-29	-28	40	40
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Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
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Axial Membrane (P)	1	1	1	1
Axial Bending (P)	27	-27	27	-27
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-25	-25	25	25
Axial Bending (ML)	-151	151	151	-151
Axial Pressure Stress	0	0	0	0

Axial Stress Summary	-148	100	204	-151
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Nozzle Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
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(Shear stress)	Outside	Inside	Outside	Inside
Shear due to (Vc)	2	2	-2	-2
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	13	13	13	13
Shear Stress Summary	15	15	10	10

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	150	132	205	192

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	4	4	4	4
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-12	-12	12	12
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	0	0	0	0
Hoop Stress Summary	-6	-6	17	17

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	1	1	1	1
Axial Bending (P)	27	-27	27	-27
Axial Membrane (Mc)	-19	-19	19	19
Axial Bending (Mc)	-218	218	218	-218
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	0	0	0	0
Axial Stress Summary	-208	173	265	-224

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-2	-2	2	2
Shear due to Torsion	13	13	13	13
Shear Stress Summary	10	10	15	15

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	209	180	266	244

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	3	3	3	3	3	3	3	3
Circ. Pl (SUS)	-30	-30	38	38	-17	-17	23	23



Circ. Q (SUS)	-58	58	82	-82	-218	218	286	-286
Long. Pm (SUS)	1	1	1	1	1	1	1	1
Long. P1 (SUS)	-14	-14	20	20	-8	-8	16	16
Long. Q (SUS)	-150	150	218	-218	-85	85	109	-109
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear P1 (SUS)	2	2	-2	-2	-2	-2	2	2
Shear Q (SUS)	12	12	12	12	12	12	12	12
Pm (SUS)	3.0	3.0	3.0	3.0	3.0	3.0	3.0	3.0
Pm+P1 (SUS)	27.3	27.3	41.2	41.2	14.5	14.5	26.4	26.4
Pm+P1+Q (Total)	165.4	138.8	239.9	197.6	232.7	204.8	313.0	261.2

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	3.00	138.00	Passed
Pm+P1 (SUS)	41.20	207.00	Passed
Pm+P1+Q (TOTAL)	313.05	413.99	Passed

Because only sustained loads were specified, the Pm+P1+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	0	0	0	0	0	0	0	0
Circ. P1 (SUS)	-30	-30	38	38	-8	-8	16	16
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	0	0	0	0	0	0	0	0
Long. P1 (SUS)	-24	-24	26	26	-18	-18	20	20
Long. Q (SUS)	-124	124	178	-178	-191	191	245	-245
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear P1 (SUS)	2	2	-2	-2	-2	-2	2	2
Shear Q (SUS)	13	13	13	13	13	13	13	13
Pm (SUS)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Pm+P1 (SUS)	30.6	30.6	38.3	38.3	18.4	18.4	20.8	20.8
Pm+P1+Q (Total)	149.9	133.4	204.7	191.3	209.6	182.3	265.9	242.9

Nozzle Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	0.00	118.00	Passed
Pm+P1 (SUS)	38.32	177.00	Passed
Pm+P1+Q (TOTAL)	265.90	353.99	Passed

Because only sustained loads were specified, the Pm+P1+Q allowable was 3 * Smh.



Input, Nozzle Desc: N26"

From: 20

Pressure for Reinforcement Calculations	P	0.1009	MPa
Temperature for Internal Pressure	Temp	65	°C
Design External Pressure	Pext	0.00	MPa
Temperature for External Pressure	Tempex	65	°C

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Cylindrical Shell	D	600.00	mm
Design Length of Section	L	657.0000	mm
Shell Finished (Minimum) Thickness	t	16.0000	mm
Shell Internal Corrosion Allowance	c	6.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		468.00	mm
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User Entered Minimum Design Metal Temperature		11.40	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-312 TP304L	
Material UNS Number		S30403	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	105.00	MPa
Allowable Stress At Ambient	Sna	115.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		6.0000	in.

Size and Thickness Basis		Nominal	
Nominal Thickness		120	

Flange Material		SA-182 F304L	
Flange Type		Weld Neck Flange	

Corrosion Allowance	can	6.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	112.6240	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	10.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		300	
Flange Grade		GR 2.3	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: N26"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation		5.501	in.
Actual Thickness Used in Calculation		0.562	in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 = $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 = $0.1 \cdot 306.0 / (138 \cdot 1.0 - 0.6 \cdot 0.1)$
 = 0.2239 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 0.1 \cdot 75.86 / (105 \cdot 1.0 - 0.6 \cdot 0.1)$
 $= 0.0730$ mm

Required Nozzle thickness under External Pressure per UG-28 : 0.0556 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 303.4508 mm
 Parallel to Vessel Wall, opening length d 151.7254 mm
 Normal to Vessel Wall (Thickness Limit), no pad Tlnp 20.6870 mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 105.0 / 138.0)$
 $= 0.761$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 105.0 / 138.0)$
 $= 0.761$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.761, 1.0)$
 $= 0.761$

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	34.852	18.893	NA
Area in Shell	A1	1444.602	1441.817	NA
Area in Nozzle Wall	A2	258.202	258.749	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds A41+A42+A43		76.089	76.089	NA
Area in Element	A5	0.000	0.000	NA
TOTAL AREA AVAILABLE	Atot	1778.893	1776.655	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (151.7254 \cdot 0.2239 \cdot 1.0 + 2 \cdot 8.2748 \cdot 0.2239 \cdot 1.0 \cdot (1 - 0.76))$
 $= 34.852$ mm²

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 151.725(1.0 \cdot 10.0 - 1.0 \cdot 0.224) - 2 \cdot 8.275$
 $(1.0 \cdot 10.0 - 1.0 \cdot 0.2239) \cdot (1 - 0.761)$
 $= 1444.602$ mm²

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp)(tn - trn)fr2$
 $= (2 \cdot 20.69)(8.27 - 0.07) \cdot 0.7609$
 $= 258.202$ mm²

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi - can / 0.707)^2 \cdot fr2$
 $= 10.0^2 \cdot 0.7609 + (0.0)^2 \cdot 0.7609$
 $= 76.089$ mm²

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $t_a = 6.0730$ mm
 Wall Thickness per UG16(b), $tr_{16b} = 7.5000$ mm
 Wall Thickness, shell/head, internal pressure $trb_1 = 6.2239$ mm
 Wall Thickness $tb_1 = \max(trb_1, tr_{16b}) = 7.5000$ mm
 Wall Thickness $tb_2 = \max(trb_2, tr_{16b}) = 7.5000$ mm
 Wall Thickness per table UG-45 $tb_3 = 12.2200$ mm

Determine Nozzle Thickness candidate [tb]:

$= \min[tb_3, \max(tb_1, tb_2)]$
 $= \min[12.22, \max(7.5, 7.5)]$
 $= 7.5000$ mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$= \max(t_a, t_b)$
 $= \max(6.073, 7.5)$
 $= 7.5000$ mm

Available Nozzle Neck Thickness $= 0.875 * 14.275 = 12.490$ mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	27.8,	Allowable	:	105.0 MPa	Passed
Expansion	:	0.0,	Allowable	:	247.2 MPa	Passed
Occasional	:	0.4,	Allowable	:	139.7 MPa	Passed
Shear	:	19.3,	Allowable	:	73.5 MPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell Weld for Nozzle skipped as Nozzle is not a Carbon Steel material.

Weld Size Calculations, Description: N26"

Intermediate Calc. for nozzle/shell Welds $T_{min} = 8.2748$ mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$5.7924 = 0.7 * t_{min}$	$7.0700 = 0.7 * W_o$ mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$= \max(0, (A-A_1+2 \cdot t_n \cdot fr_1 \cdot (E_1 \cdot t - tr)) \cdot S_v)$
 $= \max(0, (34.8522 - 1444.6016 + 2 \cdot 8.2748 \cdot 0.7609 \cdot (1.0 \cdot 10.0 - 0.2239)) \cdot 138)$
 $= \max(0, -177538.59)$ N

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$= (A_2+A_5+A_4-(W_i-Can/.707)^2 \cdot fr_2) \cdot S_v$
 $= (258.2025 + 0.0 + 76.0889 - 0.0 \cdot 0.76) \cdot 138$
 $= 46127.45$ N

Weld Load [W2]:

$= (A_2 + A_3 + A_4 + (2 \cdot t_n \cdot t \cdot fr_1)) \cdot S_v$
 $= (258.2025 + 0.0 + 76.0889 + (125.924)) \cdot 138$
 $= 63503.18$ N

Weld Load [W3]:

$= (A_2+A_3+A_4+A_5+(2 \cdot t_n \cdot t \cdot fr_1)) \cdot S$
 $= (258.2025 + 0.0 + 76.0889 + 0.0 + (125.924)) \cdot 138$
 $= 63503.18$ N

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.1416/2.0) \cdot 168.275 \cdot 10.0 \cdot 0.49 \cdot 105$
 $= 135985. \text{ N}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n$
 $= (3.1416 \cdot 80.0) \cdot (14.2748 - 6.0) \cdot 0.7 \cdot 105$
 $= 152845. \text{ N}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot D_{lo} \cdot (T - cas) \cdot 0.74 \cdot S_{ng}$
 $= (3.1416/2.0) \cdot 168.275 \cdot (16.0 - 6.0) \cdot 0.74 \cdot 138$
 $= 269902. \text{ N}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (SONW + SNW) = (135985 + 152845) = 288830 \text{ N}$

Failure Path [Path 2-2]:
 $= (Sonw + Tpgw + Tngw + Sinw)$
 $= (135985 + 0 + 269902 + 0) = 405887 \text{ N}$

Failure Path [Path 3-3]:
 $= (Sonw + Tngw + Sinw)$
 $= (135985 + 269902 + 0) = 405887 \text{ N}$

Summary of Failure Path Calculations:

Path 1-1 = 288830 N	, must exceed W = 0 N	or W1 = 46127 N
Path 2-2 = 405886 N	, must exceed W = 0 N	or W2 = 63503 N
Path 3-3 = 405886 N	, must exceed W = 0 N	or W3 = 63503 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 2 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0.001 MPa

The Drop for this Nozzle is : 12.0402 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 140.6642 mm

Input Echo, WRC297 Item 1, Description: N26" :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	6.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	600.000	mm
Shell Thickness	16.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc	138.00 MPa
Vessel Hot S.I. Allowable	Smh	138.00 MPa

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	6.0000	mm
Nozzle Diameter	139.725	mm
Nozzle Thickness	14.2748	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-312 TP304L	
Nozzle Cold S.I. Allowable	SNmc	115.00 MPa
Nozzle Hot S.I. Allowable	SNmh	105.00 MPa

Note:

*External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSstained loads.*

Design Internal Pressure	Dp	0.10	MPa
Radial Load	P	-4000.00	N
Circumferential Shear	Vc	6000.00	N
Longitudinal Shear	Vl	6000.00	N
Circumferential Moment	Mc	0.270E+07	N-mm
Longitudinal Moment	Ml	0.360E+07	N-mm
Torsional Moment	Mt	0.450E+07	N-mm

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$$

$$= 168.275 + 2 \cdot 1.65 \cdot \sqrt{(311.0(16.0 - 6.0))}$$

$$= 352.307 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	622.000 mm
Nozzle Outside Diameter	(d) =	168.275 mm
Vessel Thickness used	(T) =	10.000 mm
Nozzle Thickness used	(t) =	8.275 mm
T / t	=	1.208
d / t	=	20.336

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 2.134$$

Nr/P	=	0.059
Mr/P	=	0.108
M0/P	=	0.041
N0/P	=	0.120
MrD/Mc	=	0.237
NrDL/Mc	=	0.106
M0d/Mc	=	0.099
N0DL/Mc	=	0.112
MrD/Ml	=	0.118
NrDL/Ml	=	0.060
M0D/Ml	=	0.049
N0DL/Ml	=	0.183

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	4	4	4	4
Outplane Bending (P)	9	-9	9	-9
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-39	-39	39	39
Outplane Bending (ML)	-63	63	63	-63
Normal Pressure Stress	3	3	3	3
Outplane Stress Summary	-84	22	119	-25

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	25	-25	25	-25



Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-12	-12	12	12
Inplane Bending (ML)	-151	151	151	-151
Inplane Pressure Stress	1	1	1	1

Inplane Stress Summary	-134	116	194	-160
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Vessel Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Outplane Shear (Vc)	2	2	-2	-2
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	10	10	10	10

Shear Stress Summary	12	12	7	7
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Vessel Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Two * Max Shear Stress	137	118	194	160
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Membrane (P)	4	4	4	4
Outplane Bending (P)	9	-9	9	-9
Outplane Membrane (Mc)	-17	-17	17	17
Outplane Bending (Mc)	-95	95	95	-95
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	1	1	1	1

Outplane Stress Summary	-97	73	129	-80
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	25	-25	25	-25
Inplane Membrane (Mc)	-16	-16	16	16
Inplane Bending (Mc)	-228	228	228	-228
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	3	3	3	3

Inplane Stress Summary	-214	191	276	-231
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-2	-2	2	2
Torsional Shear (Mt)	10	10	10	10

Shear Stress Summary	7	7	12	12
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities)	Left	Left	Right	Right



	Outside	Inside	Outside	Inside
Two * Max Shear Stress	214	191	277	232

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	4	4	4	4
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-39	-39	39	39
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	1	1	1	1
Hoop Stress Summary	-33	-33	44	45

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	27	-27	27	-27
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-22	-22	22	22
Axial Bending (ML)	-164	164	164	-164
Axial Pressure Stress	0	0	0	0
Axial Stress Summary	-158	115	216	-167

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	2	2	-2	-2
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	12	12	12	12
Shear Stress Summary	14	14	9	9

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	160	151	216	213

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	4	4	4	4
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-17	-17	17	17
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	1	1	1	1
Hoop Stress Summary	-12	-11	23	23

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Axial Membrane (P)	0	0	0	0
Axial Bending (P)	27	-27	27	-27
Axial Membrane (Mc)	-17	-17	17	17
Axial Bending (Mc)	-259	259	259	-259
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	0	0	0	0

Axial Stress Summary	-247	216	305	-268
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Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-2	-2	2	2
Shear due to Torsion	12	12	12	12
Shear Stress Summary	9	9	14	14

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	247	229	306	293

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	3	3	3	3	3	3	3	3
Circ. Pl (SUS)	-35	-35	43	43	-14	-14	18	18
Circ. Q (SUS)	-54	54	72	-72	-203	203	253	-253
Long. Pm (SUS)	1	1	1	1	1	1	1	1
Long. Pl (SUS)	-10	-10	14	14	-13	-13	21	21
Long. Q (SUS)	-126	126	176	-176	-86	86	104	-104
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	2	2	-2	-2	-2	-2	2	2
Shear Q (SUS)	10	10	10	10	10	10	10	10
Pm (SUS)	3.0	3.0	3.0	3.0	3.0	3.0	3.0	3.0
Pm+Pl (SUS)	32.2	32.2	46.1	46.1	13.6	13.6	23.6	23.6
Pm+Pl+Q (Total)	137.8	118.5	191.9	161.5	214.5	192.5	275.0	233.0

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	3.00	138.00	Passed
Pm+Pl (SUS)	46.13	207.00	Passed
Pm+Pl+Q (TOTAL)	274.97	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int.	Stress Values at
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Location	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	1	1	1	1	1	1	1	1
Circ. Pl (SUS)	-35	-35	43	43	-13	-13	21	21
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	0	0	0	0	0	0	0	0
Long. Pl (SUS)	-22	-22	22	22	-17	-17	17	17
Long. Q (SUS)	-137	137	191	-191	-232	232	286	-286
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	2	2	-2	-2	-2	-2	2	2
Shear Q (SUS)	12	12	12	12	12	12	12	12
Pm (SUS)	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0
Pm+Pl (SUS)	34.3	34.3	44.2	44.2	17.7	17.7	22.7	22.7
Pm+Pl+Q (Total)	160.5	151.6	213.6	213.9	249.4	227.9	303.7	292.3

Nozzle Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	1.00	105.00	Passed
Pm+Pl (SUS)	44.18	157.50	Passed
Pm+Pl+Q (TOTAL)	303.70	315.00	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Input, Nozzle Desc: N32"

From: 20

Pressure for Reinforcement Calculations	P	0.1055	MPa
Temperature for Internal Pressure	Temp	65	°C
Design External Pressure	Pext	0.00	MPa
Temperature for External Pressure	Tempex	65	°C

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Cylindrical Shell	D	600.00	mm
Design Length of Section	L	657.0000	mm
Shell Finished (Minimum) Thickness	t	10.0000	mm
Shell Internal Corrosion Allowance	c	6.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		218.00	mm
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User Entered Minimum Design Metal Temperature		11.40	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		2.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	13.4620	mm

Flange Material		SA-182 F304L	
Flange Type		Long Weld Neck	

Corrosion Allowance	can	6.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	229.4260	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	10.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	138.00	MPa
Pad Allowable Stress At Ambient	Spa	138.00	MPa
Diameter of Pad along vessel surface	Dp	178.0000	mm
Thickness of Pad	te	10.0000	mm
Weld leg size between Pad and Shell	Wp	8.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	10.0000	mm
Reinforcing Pad Width		50.1380	mm

Flange Class		300	
Flange Grade		GR 2.3	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: N32"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation		2.000	in.
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Actual Thickness Used in Calculation 0.530 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 0.11 \cdot 306.0 / (138 \cdot 1.0 - 0.6 \cdot 0.11)$
 $= 0.2340$ mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 0.11 \cdot 31.4 / (118 \cdot 1.0 - 0.6 \cdot 0.11)$
 $= 0.0281$ mm

Required Nozzle thickness under External Pressure per UG-28 : 0.0459 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	125.6000	mm
Parallel to Vessel Wall, opening length	d	62.8000	mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		10.0000	mm

The Pad diameter is greater than the Diameter Limit. The excess will not be considered.

*Taking a UG-36(c)(3)(a) exemption for nozzle: N3.2".
 This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 6.0459	mm
Wall Thickness per UG16(b),	tr16b	= 7.5000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 6.2340	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 7.5000	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 7.5000	mm
Wall Thickness per table UG-45	tb3	= 10.8000	mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[10.8, \max(7.5, 7.5)]$
 $= 7.5000$ mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(6.0459, 7.5)$
 $= 7.5000$ mm

Available Nozzle Neck Thickness = 13.4620 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	18.3,	Allowable	:	118.0 MPa	Passed
Expansion	:	0.0,	Allowable	:	276.7 MPa	Passed
Occasional	:	0.2,	Allowable	:	156.9 MPa	Passed
Shear	:	14.7,	Allowable	:	82.6 MPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govern. thk, tg = 10.0, c = 6.0 mm, E* = 1.0
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.059$, Temp. Reduction = 78 °C
 Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C



Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, tg = 10.0, c = 6.0 mm, E* = 1.0
 Thickness Ratio = $tr * E*/(tg - c) = 0.059$, Temp. Reduction = 78 °C
 Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, tg = 10.0, tr = 0.234, c = 6.0 mm, E* = 1.0
 Thickness Ratio = $tr * E*/(tg - c) = 0.059$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, tg = 10.0, tr = 0.234, c = 6.0 mm, E* = 1.0
 Thickness Ratio = $tr * E*/(tg - c) = 0.059$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Gov. MDMT of the Nozzle : -104 °C
 Gov. MDMT of the Reinforcement Pad : -104 °C
 Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Weld Size Calculations, Description: N32"

Intermediate Calc. for nozzle/shell Welds Tmin 7.4620 mm
 Intermediate Calc. for pad/shell Welds TminPad 4.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$5.2234 = 0.7 * tmin.$	$7.0700 = 0.7 * Wo$ mm
Pad Weld	$2.0000 = 0.5 * TminPad$	$5.6560 = 0.7 * Wp$ mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 2.5278 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 241.9537 mm

Input Echo, WRC297 Item 1, Description: N32" :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	6.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	600.000	mm
Shell Thickness	10.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc	138.00 MPa
Vessel Hot S.I. Allowable	Smh	138.00 MPa

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	6.0000	mm
Nozzle Diameter	50.800	mm
Nozzle Thickness	13.4620	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-106 B	
Nozzle Cold S.I. Allowable	SNmc	118.00 MPa
Nozzle Hot S.I. Allowable	SNmh	118.00 MPa

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter of Reinforcing Pad 178.000 mm
 Thickness of Reinforcing Pad 10.0000 mm

Note:

*External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSTAINED loads.*

Design Internal Pressure	Dp	0.11	MPa
Radial Load	P	-1300.00	N
Circumferential Shear	Vc	2000.00	N
Longitudinal Shear	Vl	2000.00	N
Circumferential Moment	Mc	300000.00	N-mm
Longitudinal Moment	Ml	400000.03	N-mm
Torsional Moment	Mt	500000.00	N-mm

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

*Warning - The ratio, Dn/Tn (5.774) is < 10
 check the Limitations of WRC bulletin 297.*

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(\text{t} - \text{ca}))}$$

$$= 77.724 + 2 \cdot 1.65 \cdot \sqrt{(313.0(10.0 - 6.0))}$$

$$= 194.490 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	626.000 mm
Nozzle Outside Diameter	(d) =	77.724 mm
Vessel Thickness used	(T) =	14.000 mm
Nozzle Thickness used	(t) =	7.462 mm
T / t	=	1.876
d / t	=	10.416

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.830$$

Nr/P	=	0.152
Mr/P	=	0.147
M0/P	=	0.077
N0/P	=	0.148
MrD/Mc	=	0.226
NrDL/Mc	=	0.186
M0d/Mc	=	0.151
N0DL/Mc	=	0.115
MrD/Ml	=	0.182
NrDL/Ml	=	0.155
M0D/Ml	=	0.122
N0DL/Ml	=	0.124

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	0	0	0	0
Outplane Bending (P)	3	-3	3	-3
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (Ml)	-3	-3	3	3
Outplane Bending (Ml)	-19	19	19	-19
Normal Pressure Stress	2	2	2	2



Outplane Stress Summary -16 16 28 -15

Vessel Stresses (MPa):

LONGITUDINAL PLANE
 (Stresses parallel to longitudinal plane)

	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
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Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	5	-5	5	-5
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-4	-4	4	4
Inplane Bending (ML)	-28	28	28	-28
Inplane Pressure Stress	1	1	1	1

Inplane Stress Summary -24 20 40 -28

Vessel Stresses (MPa):

LONGITUDINAL PLANE
 (Shear stress normal to longitudinal plane)

	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
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Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	3	3	3	3

Shear Stress Summary 4 4 2 2

Vessel Stresses (MPa):

LONGITUDINAL PLANE
 (Stress Intensities)

	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
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Two * Max Shear Stress	27	24	41	28
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE
 (Stresses Normal to circumferential plane)

	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Outplane Membrane (P)	0	0	0	0
Outplane Bending (P)	3	-3	3	-3
Outplane Membrane (Mc)	-2	-2	2	2
Outplane Bending (Mc)	-17	17	17	-17
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	1	1	1	1

Outplane Stress Summary -14 14 25 -16

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE
 (Stresses parallel to circumferential plane)

	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	5	-5	5	-5
Inplane Membrane (Mc)	-3	-3	3	3
Inplane Bending (Mc)	-26	26	26	-26
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	2	2	2	2

Inplane Stress Summary -21 20 39 -25

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE
 (Shear stress normal to circumferential plane)

	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	3	3	3	3
Shear Stress Summary	2	2	4	4

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	22	21	41	27

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	0	0	0	0
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-3	-3	3	3
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	0	0	0	0
Hoop Stress Summary	-1	-1	4	4

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	9	-9	9	-9
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-15	-15	15	15
Axial Bending (ML)	-58	58	58	-58
Axial Pressure Stress	0	0	0	0
Axial Stress Summary	-62	34	84	-51

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	2	2	-2	-2
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	7	7	7	7
Shear Stress Summary	9	9	4	4

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	63	40	84	57

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	0	0	0	0
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-2	-2	2	2
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0



Hoop Pressure Stress	0	0	0	0
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Hoop Stress Summary	0	0	3	3
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Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Axial Membrane (P)	0	0	0	0
Axial Bending (P)	9	-9	9	-9
Axial Membrane (Mc)	-11	-11	11	11
Axial Bending (Mc)	-55	55	55	-55
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	0	0	0	0

Axial Stress Summary	-55	35	77	-52
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Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
---	---------------	--------------	---------------	--------------

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-2	-2	2	2
Shear due to Torsion	7	7	7	7

Shear Stress Summary	4	4	9	9
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Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu Outside	Cl Inside	Du Outside	Dl Inside
--	---------------	--------------	---------------	--------------

Two * Max Shear Stress	56	37	78	59
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Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$$

$$= 77.724 + 2 \cdot 1.65 \cdot \sqrt{(308.0(10.0 - 6.0))}$$

$$= 193.554 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 616.000 mm

Nozzle Outside Diameter (d) = 178.000 mm

Vessel Thickness used (T) = 4.000 mm

Nozzle Thickness used (t) = 57.600 mm

T / t = 0.069

d / t = 10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 3.586$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P = 0.057

Mr/P = 0.093

M0/P = 0.026

N0/P = 0.031

MrD/Mc = 0.258

NrDL/Mc = 0.111

M0d/Mc = 0.077

N0DL/Mc = 0.054

MrD/Ml = 0.066

NrDL/Ml = 0.072

M0D/Ml = 0.020

N0DL/Ml = 0.038

Vessel Stresses (MPa):



LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	2	2	2	2
Outplane Bending (P)	12	-12	12	-12
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-5	-5	5	5
Outplane Bending (ML)	-16	16	16	-16
Normal Pressure Stress	8	8	8	8
Outplane Stress Summary	1	9	45	-13

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	4	4	4	4
Inplane Bending (P)	45	-45	45	-45
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-10	-10	10	10
Inplane Bending (ML)	-55	55	55	-55
Inplane Pressure Stress	4	4	4	4
Inplane Stress Summary	-11	8	119	-82

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	2	2	2	2
Shear Stress Summary	4	4	0	0

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	15	13	119	82

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	2	2	2	2
Outplane Bending (P)	12	-12	12	-12
Outplane Membrane (Mc)	-5	-5	5	5
Outplane Bending (Mc)	-48	48	48	-48
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	4	4	4	4
Outplane Stress Summary	-35	36	73	-49

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)	4	4	4	4
Inplane Bending (P)	45	-45	45	-45
Inplane Membrane (Mc)	-11	-11	11	11



Inplane Bending (Mc)	-163	163	163	-163
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	8	8	8	8

Inplane Stress Summary	-117	119	232	-184
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	2	2	2	2

Shear Stress Summary	0	0	4	4
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Two * Max Shear Stress	117	119	232	184

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	2	2	2	2	2	2	2	2
Circ. Pl (SUS)	-3	-3	3	3	-2	-2	4	4
Circ. Q (SUS)	-16	16	22	-22	-21	21	31	-31
Long. Pm (SUS)	1	1	1	1	1	1	1	1
Long. Pl (SUS)	-3	-3	5	5	-2	-2	2	2
Long. Q (SUS)	-23	23	33	-33	-14	14	20	-20
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	3	3	3	3	3	3	3	3
Pm (SUS)	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
Pm+Pl (SUS)	2.6	2.6	6.6	6.6	2.2	2.2	6.3	6.3
Pm+Pl+Q (Total)	26.7	23.0	39.3	27.4	21.6	21.5	38.1	26.7

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	2.00	138.00	Passed
Pm+Pl (SUS)	6.62	207.00	Passed
Pm+Pl+Q (TOTAL)	39.32	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl



Circ. Pm (SUS)	8	8	8	8	8	8	8	8
Circ. Pl (SUS)	-3	-3	7	7	-7	-7	15	15
Circ. Q (SUS)	-4	4	28	-28	-118	118	208	-208
Long. Pm (SUS)	4	4	4	4	4	4	4	4
Long. Pl (SUS)	-6	-6	14	14	-3	-3	7	7
Long. Q (SUS)	-10	10	100	-100	-36	36	60	-60
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	2	2	2	2	2	2	2	2
Pm (SUS)	8.0	8.0	8.0	8.0	8.0	8.0	8.0	8.0
Pm+Pl (SUS)	7.3	7.3	18.3	18.3	2.0	2.0	23.1	23.1
Pm+Pl+Q (Total)	14.3	11.5	118.0	82.0	117.0	119.0	231.1	185.1

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	8.00	138.00	Passed
Pm+Pl (SUS)	23.08	207.00	Passed
Pm+Pl+Q (TOTAL)	231.06	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	0	0	0	0	0	0	0	0
Circ. Pl (SUS)	-3	-3	3	3	-2	-2	2	2
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	0	0	0	0	0	0	0	0
Long. Pl (SUS)	-15	-15	15	15	-11	-11	11	11
Long. Q (SUS)	-49	49	67	-67	-46	46	64	-64
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	2	2	-2	-2	-2	-2	2	2
Shear Q (SUS)	7	7	7	7	7	7	7	7
Pm (SUS)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Pm+Pl (SUS)	15.3	15.3	15.3	15.3	11.4	11.4	11.4	11.4
Pm+Pl+Q (Total)	65.3	41.1	82.3	55.9	57.5	38.3	76.1	57.9

Nozzle Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	0.00	118.00	Passed
Pm+Pl (SUS)	15.32	177.00	Passed
Pm+Pl+Q (TOTAL)	82.32	353.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.



Input, Nozzle Desc: N42"

From: 40

Pressure for Reinforcement Calculations	P	0.1000	MPa
Temperature for Internal Pressure	Temp	65	°C
Design External Pressure	Pext	0.00	MPa
Temperature for External Pressure	Tempex	65	°C

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Outside Diameter of Bolted Blind Flange	D	771.00	mm
Head Finished (Minimum) Thickness	t	38.0000	mm
Head Internal Corrosion Allowance	c	6.0000	mm
Head External Corrosion Allowance	co	0.0000	mm

Distance from Head Centerline	L1	0.0000	mm
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User Entered Minimum Design Metal Temperature		11.40	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		2.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	13.4620	mm

Flange Material		SA-105	
Flange Type		Long Weld Neck	

Corrosion Allowance	can	6.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	229.4260	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	38.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	138.00	MPa
Pad Allowable Stress At Ambient	Spa	138.00	MPa
Diameter of Pad along vessel surface	Dp	100.0000	mm
Thickness of Pad	te	38.0000	mm
Weld leg size between Pad and Shell	Wp	30.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	38.0000	mm
Reinforcing Pad Width		11.1380	mm

Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: N42"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation		2.000	in.
Actual Thickness Used in Calculation		0.530	in.

Note:

Post Weld Heat Treatment is required for this nozzle and it was not specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 0.1 \cdot 31.4 / (118 \cdot 1.0 - 0.6 \cdot 0.1)$
 $= 0.0266 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.0459 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 141.7240 mm
 Parallel to Vessel Wall Rn+tn+t 70.8620 mm
 Normal to Vessel Wall (Thickness Limit), pad side Tlwp 37.9466 mm

Weld Strength Reduction Factor [fr1]:

$= \min(1, S_n / S_v)$
 $= \min(1, 118.0 / 138.0)$
 $= 0.855$

Weld Strength Reduction Factor [fr2]:

$= \min(1, S_n / S_v)$
 $= \min(1, 118.0 / 138.0)$
 $= 0.855$

Weld Strength Reduction Factor [fr4]:

$= \min(1, S_p / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:

$= \min(fr2, fr4)$
 $= \min(0.855, 1.0)$
 $= 0.855$

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	950.018	950.251	NA
Area in Shell	A1	211.244	210.694	NA
Area in Nozzle Wall	A2	482.513	481.260	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds A41+A42+A43		816.497	816.497	NA
Area in Element	A5	429.740	429.740	NA
TOTAL AREA AVAILABLE	Atot	1939.993	1938.191	NA

The External Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.

The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness
 Based on given Pad Thickness: 77.7240 38.0000 mm
 Based on given Pad Diameter: 100.0000 0.0000 mm
 Based on Shell or Nozzle Thickness: 77.7240 13.4620 mm

Area Required [A]:

$= 0.5 \cdot d \cdot t + t \cdot t_n (1 - fr1)$ per UG-39(b)(1)
 $= 0.5 \cdot 62.8 \cdot 29.2552 + 29.2552 \cdot 7.462 (1 - 0.86)$
 $= 950.251 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 78.924(1.0 \cdot 32.0 - 1.0 \cdot 29.255) - 2 \cdot 7.462$$

$$(1.0 \cdot 32.0 - 1.0 \cdot 29.2552) \cdot (1 - 0.855)$$

$$= 210.694 \text{ mm}^2$$

Area Available in Nozzle Wall Projecting Outward [A2]:

$$= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$$

$$= (2 \cdot 37.95) \cdot (7.46 - 0.05) \cdot 0.8551$$

$$= 481.260 \text{ mm}^2$$

Area Available in Welds [A41 + A42 + A43]:

$$= (Wo^2 - Ar \text{ Lost}) \cdot Fr3 + ((Wi - can/0.707)^2 - Ar \text{ Lost}) \cdot fr2 + Trapfr4$$

$$= (0.0) \cdot 0.86 + (0.0) \cdot 0.86 + 816.4971^2 \cdot 1.0$$

$$= 816.497 \text{ mm}^2$$

Area Available in Element [A5]:

Pad Thickness is limited by 30 degree angle to 19.2916 mm

$$= (\min(Dp, DL) - (\text{Nozzle OD})) \cdot (\min(tp, Tlwp, te)) \cdot fr4$$

$$= (100.0 - 77.724) \cdot 19.2916 \cdot 1.0$$

$$= 429.740 \text{ mm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 6.0459 mm
Wall Thickness per UG16(b),	tr16b = 7.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 35.2480 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 35.2480 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 7.5000 mm
Wall Thickness per table UG-45	tb3 = 10.8000 mm

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[10.8, \max(35.248, 7.5)]$$

$$= 10.8000 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(6.0459, 10.8)$$

$$= 10.8000 \text{ mm}$$

Available Nozzle Neck Thickness = 13.4620 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, tg = 13.462, tr = 0.027, c = 6.0 mm, E* = 1.0
 Thickness Ratio = tr * E*/(tg - c) = 0.004, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-19 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, tg = 13.462, tr = 0.027, c = 6.0 mm, E* = 1.0
 Thickness Ratio = tr * E*/(tg - c) = 0.004, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-19 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, tg = 38.0, tr = 0.279, c = 6.0 mm, E* = 1.0
 Thickness Ratio = tr * E*/(tg - c) = 0.009, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	10 °C
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Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, tg = 13.462, tr = 0.027, c = 6.0 mm, E* = 1.0
 Thickness Ratio = tr * E*/(tg - c) = 0.004, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -19 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Gov. MDMT of the Nozzle : -104 °C
 Gov. MDMT of the Reinforcement Pad : -104 °C
 Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = 0.10/5.10 = 0.020

Weld Size Calculations, Description: N42"

Intermediate Calc. for nozzle/shell Welds Tmin 7.4620 mm
 Intermediate Calc. for pad/shell Welds TminPad 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	5.2234 = 0.7 * tmin.	7.0700 = 0.7 * Wo mm
Pad Weld	9.5000 = 0.5*TminPad	21.2100 = 0.7 * Wp mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot tn \cdot fr1 \cdot (E1 \cdot t - tr)) \cdot Sv)$
 $= \max(0, (950.2512 - 210.6939 + 2 \cdot 7.462 \cdot 0.8551 \cdot (1.0 \cdot 32.0 - 29.2552)) \cdot 138)$
 $= 106881.58 \text{ N}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(Wi-Can/.707)^2 \cdot fr2) \cdot Sv$
 $= (481.2602 + 429.7395 + 816.4971 - 0.0 \cdot 0.86) \cdot 138$
 $= 238369.97 \text{ N}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot tn \cdot t \cdot fr1)) \cdot Sv$
 $= (481.2602 + 0.0 + 85.5072 + (408.3553)) \cdot 138$
 $= 134553.06 \text{ N}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot tn \cdot t \cdot fr1)) \cdot S$
 $= (481.2602 + 0.0 + 816.4971 + 429.7395 + (408.3553)) \cdot 138$
 $= 294717.19 \text{ N}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$
 $= (3.1416/2.0) \cdot 77.724 \cdot 10.0 \cdot 0.49 \cdot 118$
 $= 70584. \text{ N}$

Shear, Pad Element Weld [Spew]:
 $= (\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW$
 $= (3.1416/2.0) \cdot 100.0 \cdot 30.0 \cdot 0.49 \cdot 138$
 $= 318619. \text{ N}$

Shear, Nozzle Wall [Snw]:

$$\begin{aligned} &= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn \\ &= (3.1416 \cdot 35.131) \cdot (13.462 - 6.0) \cdot 0.7 \cdot 118 \\ &= 68019. \text{ N} \end{aligned}$$

Tension, Pad Groove Weld [Tpgw]:

$$\begin{aligned} &= (\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg \\ &= (3.1416/2) \cdot 77.724 \cdot 38.0 \cdot 0.74 \cdot 138 \\ &= 473723. \text{ N} \end{aligned}$$

Tension, Shell Groove Weld [Tngw]:

$$\begin{aligned} &= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng \\ &= (3.1416/2.0) \cdot 77.724 \cdot (38.0 - 6.0) \cdot 0.74 \cdot 138 \\ &= 398925. \text{ N} \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SPEW + SNW) = (318619 + 68019) = 386638 \text{ N}$$

Failure Path [Path 2-2]:

$$\begin{aligned} &= (Sonw + Tpgw + Tngw + Sinw) \\ &= (70584 + 473723 + 398925 + 0) = 943232 \text{ N} \end{aligned}$$

Failure Path [Path 3-3]:

$$\begin{aligned} &= (Spew + Tngw + Sinw) \\ &= (318619 + 398925 + 0) = 717544 \text{ N} \end{aligned}$$

Summary of Failure Path Calculations:

Path 1-1 = 386638 N	, must exceed W = 106881 N	or W1 = 238369 N
Path 2-2 = 943232 N	, must exceed W = 106881 N	or W2 = 134553 N
Path 3-3 = 717543 N	, must exceed W = 106881 N	or W3 = 294717 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 1 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0.001 MPa

The Cut Length for this Nozzle is, Drop + Ho + H + T : 267.4768 mm

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Noted issues for the current model:

Listed below are the known problem areas for the current design. If one or more of the design flags are turned on, please re-run the analysis. Some of these issues may be resolved when using updated input values.

**** Warning: PWHT was required for at least 1 Nozzle/Pad in this vessel!**

Please review all reports carefully!

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